Under the therapist's guidance, most clients can learn basic skills to provide effective therapy for their pet. Clients with working dogs or high-level athletes may carry out more complex home exercises to refine specific skills. Recheck evaluations are needed to ensure that the client is performing the therapies correctly and the patient is progressing toward the therapeutic goals.

## References


Bandy WD, Irion JM, and Briggler M (1997) The effect of time and frequency of static stretching on flexibility of the hamstring muscles. *Phys Ther* 77(10):1090–1096.

Cameron MH (2003) *Physical Agents in Rehabilitation: From Research to Practice*, 2nd edn. Saunders, St. Louis, MO.

Carr JG, Millis DL, and Weng HY (2013) Exercises in canine physical rehabilitation: Range of motion of the forelimb during stair and ramp ascent. *J Small Anim Pract* 54(8):409–413.

Deliagina TG, Beloozerova IN, Orlovsky GN, and Zelenin PV (2014) Contribution of supraspinal systems to generation of automatic postural responses. *Front Integr Neurosci* 8:1–20.

Dragone L, Heinrichs K, Levine D, *et al.* (2014) Superficial thermal modalities. In *Canine Rehabilitation and Physical Therapy*, 2nd edn (eds. DL Millis and D Levine). Elsevier Saunders, Philadelphia, PA, pp. 312–327.

Gaynor JS and Muir WW (2015) *Handbook of Veterinary Pain Management*, 3rd edn. Elsevier, St Louis, MO.

Kisner C and Colby LA (2007) *Therapeutic Exercise*, 5th edn. F.A. Davis Co., Philadelphia, PA.

Millard RP, Headrick JF, and Millis DL (2010) Kinematic analysis of the pelvic limbs of healthy dogs during stair and decline slope walking. *J Small Anim Pract* 51(8):419–422.

Millis DL and Levine D (2014) Range-of-motion and stretching exercises. In *Canine Rehabilitation and Physical Therapy*, 2nd edn (eds. DL Millis and D Levine). Elsevier Saunders, Philadelphia, PA, pp. 431–446.

Millis DL, Drum M, and Levine D (2014a) Therapeutic exercises: Early limb use exercises. In *Canine Rehabilitation and Physical Therapy*, 2nd edn (eds. DL Millis and D Levine). Elsevier Saunders, Philadelphia, PA, pp. 495–505.

Millis DL, Drum M, and Levine D (2014b) Therapeutic exercises: Joint motion, strengthening, endurance, and speed exercises. In *Canine Rehabilitation and Physical Therapy*, 2nd edn (eds. DL Millis and D Levine). Elsevier Saunders, Philadelphia, PA, pp. 506–525.

Page P (2012) Current concepts in muscle stretching for exercise and rehabilitation. *Int J Sports Phys Ther* 7(1):109–119.

Salvin HE, McGreevy PD, Sachdev PS, and Valenzuela MJ (2010). Under diagnosis of canine cognitive dysfunction: a cross-sectional survey of older companion dogs. *Vet J* 184(3):277–281.

Sutton A and Whitlock D (2014) Massage. In *Canine Rehabilitation and Physical Therapy*, 2nd edn (eds. DL Millis and D Levine). Elsevier Saunders, Philadelphia, PA, pp. 464–483.

# 7

# Supporting the Client and Patient

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CHAPTER MENU



*Medicine should be practiced as a form of friendship.*
Leon Bernard, French physician (early 1900s)

*We are the first generation to allow pets to sleep in our beds.*
Marty Becker, DVM, 2010 (personal conversation)

## Introduction

In recent decades, pets have made a migration of biblical proportion from the back yard to the bedroom and from the kennel to the couch. Studies consistently reveal that 80–85% of pet owners consider their pets to be members of the family. Companion animal ownership is associated with a range of physical, psychological, and social health advantages (Smith, 2012). Pet health is important to the human family. The human–animal bond refers to the attachment between a human and their pet. Attachment refers to a particular type of long-lasting affectional bond that develops between two individuals (Bowlby, 1969; Bretherton, 1992). It provides a sense of security and reduces feelings of stress and anxiety (Carter, 1998). The relationship between dog and owner/caregiver has been proposed to resemble the child–parent bond (Kubinyi *et al.*, 2003; Serpell, 2003; Fallani *et al.*,2006). Overall, studies show that pet dogs react in a similar way to infants when participating with their owners in a strange

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situation challenge (Topál *et al.*,1998; Rehn *et al.*, 2013). The relationship between cat and caregiver has some similarities, despite cat affection being difficult to prove (Vitale Shreve and Udell, 2015). The caregiver of a cat certainly has the same strong feelings that a dog owner does. Advances in veterinary medicine provide the opportunity for pets to live longer and better than ever. Physical rehabilitation has facilitated mobility and therefore healthier lives for many animals. The veterinary technician or nurse fulfills the role as advocate on behalf of beings who cannot advocate for themselves.

In order to fulfill this advocacy role, it is critical for veterinary technicians involved with providing physical rehabilitation to understand the importance of actively engaging both the client and the pet in the planned care, as well as how to accomplish that goal. Compliance is the key.

## The Effect of Pain and Morbidity

### How Pain and Dysfunction Undermine the Human–Animal Bond

Palliative care in the health care arena means to treat the symptoms of a disease without the intention of curing it. Because pain is a major factor in diminished quality of life, pain management plays a critical role in applying palliative care and hospice principles to pets (Downing, 2011a) as well as a critical role in improving function for any rehabilitation patient.

Understanding pain physiology is critical to treating painful conditions, whether end-of-life or ongoing, because it allows a layering of various chemicals and modalities to best address the needs of the individual patient. Likewise, a specific understanding of how pain is expressed, both within a species but also how it can vary between individuals, allows more direct targeting of analgesic therapy. These can all be reviewed in Chapter 3.

Teaching the client/caregivers to recognize symptoms of pain in their pet allows the client to take an active role in helping their pet cope with the stranglehold unrelieved pain can cause. Understanding the nuances of diagnosing and treating pain is even more critical in long-term and end-of-life care, because often these treatments will be fairly prolonged, and so the side-effects of each treatment may have time to become fully expressed. A host of treatments for pain are increasingly available in veterinary medicine. These include drugs, acupuncture, physical modalities, and interventional medicine.

Pain medicine, like palliative medicine, is highly tied into the social and psychological health of humans and companion animals (Wright, 2013). This complex interplay between medicine, ethics, and compassion is a daily conversation in which all three components drive decision-making, including but not limited to the application of evidence-based theory and practice. In long-term care, both acute and pathologic pain may be present, but nearly always there will be chronic pathologic pain components. It is highly recommended that a composite, behavioral type of pain scale is utilized in this setting. Visceral and deep sources of pain will often reflect onto muscles and fascia that share the neurological framework. The body can serve as a roadmap to the sources of pain and discomfort that the patient is experiencing (Wright, 2013).

There are certain aspects of pathologic pain that may be measured in animals. Depression can lead to morbidity. One of the core symptoms of depression is anhedonia, the decrease and loss of interest in pleasurable activities. For pets, we measure interest in food or motivation for play. We also measure how they are interacting socially with other animals in the group, and changes in sleep patterns and daytime activity level. Another behavior that has been used frequently to measure animal depression is whether they readily give up when exposed to a stressful situation. Clients need to be educated to watch for these behaviors in their pets during the rehabilitation process.

## Aims of Physical Rehabilitation

### The Role of Physical Rehabilitation in Managing Pain and Restoring Function

The aims and benefits of physical rehabilitation are many, including reduction of pain; promotion of the healing process; muscle strengthening, improving joint flexibility; promoting/restoring normal movement patterns; cardiovascular fitness; engaging mentally and enhancing the patient's sense of well-being (Sharp, 2008; Downing, 2011b). Rehabilitation helps an individual that has had an illness or injury to achieve the highest level of function, independence, and quality of life possible. The success of any surgery is as much down to the rehabilitation carried out as to the surgical technique performed (Sharp, 2008).

Pets tend to lose muscle and balance with age (Bellows *et al.*, 2015). This age-related muscle and proprioceptive loss can lead to inactivity and weight gain, increased stress on joints from diminished stability of surrounding musculature, and injuries from tripping and falling (Jurek and McCauley, 2011). Chronically ill patients, particularly patients experiencing chronic, pathologic (maladaptive pain), are generally less active than their non-painful counterparts. Prolonged inactivity leads to a state of deconditioning in which muscle strength is lost and endurance declines. Muscles may contract, leading to decreased range of motion (ROM), and bones may become osteopenic (Downing, 2011b). Rehabilitation supports the client and patient in the home environment because it improves the ability of the pet to perform the activities of daily living (sometimes with the help of the client) (Figure 7.1). Improving home activity and ease of motion is one of the main aims of physical rehabilitation.

## Client Engagement and Compliance

Compliance for the veterinary health care team means that pets, including those pets who can benefit from physical rehabilitation, receive the care that the veterinarian knows they need and deserve. In the past,

Figure 7.1 Improving home activity by practicing the stairs in clinic.

compliance meant asking, "Did the client do as he or she was instructed?" Because of this approach, veterinarians have consistently overestimated client compliance with medical recommendations.

The 2003 American Animal Hospital Association compliance study (AAHA, 2003) showed that the biggest single failing in achieving good compliance for pet health care was the lack of an effective recommendation. The data further identified that clients *want* what is best for their pets, and that they *would* comply with veterinary recommendations if the practice and veterinary health care team tried to assist them.

Some of the obstacles to effective recommendations include:

- inadequate communication with clients;
- inadequate education of clients about why certain treatments are necessary;
- information overload for the client;
- overestimation of the client's willingness or ability to be involved in and to pay for the pet's care.

Improving compliance consists of three important factors working together:

$$\text{Compliance} = \text{Recommendation} + \text{Acceptance} + \text{Follow-Through}$$
$$(C = R + A + FT)$$

The word "CRAFT" is an easy way to remember how to better engage the client in the pet's physical rehabilitation.

Each rehabilitation patient needs an effective recommendation from the veterinarian made with conviction, and with the specific needs of the pet at its core. The veterinary technician will play a critical role both in communicating the details of the veterinarian's recommendation as well as clarifying the client's understanding of the recommendation (Figure 7.2). The client must next accept the specific rehabilitation recommendation made on behalf of the pet. This acceptance will hinge both on the client's understanding of what will be required of them and on the logistics surrounding the rehabilitation plan itself.

Figure 7.2 Client engagement means deep conversations.

Finally, the veterinary technician plays a critical role in the necessary follow-through.

Goals include:

- ensuring that the rehabilitation patient receives the prescribed therapy by providing reminder calls, emails, or text messages, depending upon the client's preference;
- supporting the client as needed with appropriate updates about the pet's progress;
- fine-tuning the client's at home participation in rehabilitation;
- serving to reinforce the details of the veterinarian's messaging to the client about the pet's ongoing rehabilitation needs.

### What Are the Steps to Improving Compliance for Physical Rehabilitation Patients?

**Step One**

The veterinary team must speak with one voice. The veterinary technician must understand the rehabilitation treatment plan and be able to discuss the goals and details of treatment with the client. Educating the client about the rehabilitation plan, encouraging ongoing participation of the client in that plan, and reassuring the client when needed are all important roles the veterinary technician can fill. As an extension of the veterinarian, the veterinary technician facilitates continuity of care as rehabilitation progresses and once the pet "graduates" from the rehabilitation program.

**Step Two**

Answer two important questions: "How do we provide pet rehabilitation here?" and "How do we speak to clients about how we provide pet rehabilitation here?" Veterinary technicians have an opportunity and an obligation to be able to answer these two questions. By knowing the details of the treatment plan, the veterinary technician will participate in delivery of rehabilitation and also provide strong, detailed communication with the client to coordinate what happens in the facility with the rehabilitation care that occurs at home.

**Step Three**

Provide consistency and follow-through to facilitate successful rehabilitation for each patient. This means coordinating the "3 R's" of rehabilitation:

- rechecking the patient alongside the veterinarian
- reassessing the patient in dialog with the client
- revising the treatment plan and assisting with execution.

The veterinary technician should understand the arc of care and outcome expectations for each rehabilitation patient. Schedule the next reassessment and/or rehabilitation appointment before the client leaves following the current session. The veterinary technician truly serves as a facilitator of care for the rehabilitation patient, working with all three parties – the patient, the veterinarian, and the pet owner.

Success for the veterinary technician working with rehabilitation patients means leaving the client not with unanswered questions, but instead with unquestioned answers. The veterinary technician's activities during physical rehabilitation serve to enhance, lengthen, and strengthen the precious human–animal bond. There is no greater contribution for a veterinary technician to make as he or she advocates on behalf of a being who cannot advocate for itself!

## Helping the Client to Make a Decision

Many of the practices and techniques that are considered to be physical rehabilitation can be taught to the client so that they may be performed each day. Specific activities, procedures, and techniques may be performed as tolerated to avoid compromising pain management or overly fatiguing the patient. Physical medicine and rehabilitation provide non-pharmacologic interventions and tools used to enhance care of the patient in palliative and hospice care (Downing, 2011b).

The skilled technician is a source of vital information required to administer appropriate therapies that the rehabilitation veterinarian has chosen. He or she is a trusted caretaker for both hospitalized and outpatients. The success of this relationship is terribly important for all patients and applies to elective, routine, and extraordinary cases.

Client education begins at the initial visit. Providing verbal, written, and hands-on instruction is imperative for client understanding. Clients need to be taught that physical rehabilitation can extend quality of life for their pet. Owners of pets with life-limiting conditions may feel confused and helpless; increasing their involvement in the decision-making and care of their pet can provide them with a feeling of empowerment. High-quality nursing care must be provided for patients when there is no longer an expectation of achieving a cure (Kerrigan, 2013). The major goals of such care include maintaining physical comfort, including assistance with the animal's bodily functions and activities of daily living, minimizing any complications and side-effects of the disease or medication, and prevention or relief of pain and other physical discomforts including nausea, anorexia, and diarrhea. Time spent with an owner taking a "daily activity" history is beneficial for the comfort of both the patient and the owner. Providing pleasures such as companionship and social interaction, along with mental stimulation and play (permitted by the animal's physical condition) (Figure 7.3) can help greatly offset the unpleasant physical feelings associated with the disease and relative confinement (Kerrigan, 2013). In general, the animal's attitude, responsiveness, and enthusiasm for interactions with the family members must be assessed regularly. If the animal's psychosocial condition declines, it is imperative to reassess the nursing interventions, methods of analgesia, and owner activities to make any necessary adjustments that may restore quality of life. Psychosocial care also includes assistance for family members in dealing with the distress and burden of caring for their sick pet, along with other difficulties such as financial concerns or the acknowledgement of anticipatory grief (Balducci, 2007).

Figure 7.3 Providing mental stimulation and play as permitted by the animal's physical condition.

## Financial Decisions

In the veterinary world we come across many different scenarios when it comes to clients making financial decisions for their pets so it would be a mistake to generalize; however, some examples of client type are used here to illustrate barriers to compliance:

- *The price shopper* – This client values veterinary care but shows great loyalty to practitioners who demonstrate that they keep costs down. This client questions any optional treatment program, whether pain or physical rehabilitation, and asks about the need for any items that you itemize in a treatment plan. One way to talk to these clients is to break down the cost of treatment on a day-to-day basis, thereby showing that pain control and physical rehabilitation is really a very inexpensive part of the process of caring for the animal.

– *TIP*: Take the time to answer questions about the importance of pain control and physical rehabilitation.

- *The procrastinator* – This client values veterinary care and likes your clinic, but finds visits stressful and so is less likely to visit regularly. Regular clinic visits are a vital component of a comprehensive rehabilitation plan; a patient will take longer to benefit and will get less benefit from a protracted, infrequent therapy plan. Explain the risks of waiting to treat the issue and the effect of infrequent visits. Attempt to remove barriers to needed visits (pet taxi, time of day, wait time), procedures or medications with as much convenience and connection as you can muster.
  – *TIP*: Listen to the client's needs and interests. Take time to form the long-term rapport you need to bond with the client and their pet.
- *The avoider* – This client is sometimes distrustful and likely has a do-it-yourself mentality. In these cases, emphasize the urgency of performing any procedure and of providing pain control. When it comes to procedures, emphasize the long-term effects of client decisions made in the moment. A client needs to know that bad management of acute pain, including not moving forward with physical rehabilitation, can set up patients for chronic, long-term pain.
  – *TIP:* Smart practitioners can disagree about whether the highest standard is an "always" standard when well-meaning people lack funds or you cannot provide the highest care for less every time. Explore your own practice philosophy.
- *The neglector* – This client is the most passive type of pet owner, who strongly resists investment in even minimum care. But sometimes neglectors just do not know what the pet needs. Their attitude can change with the right education; we know they care about their pet.
  – *TIP*: Stop using words like "ought to" and "should" and start using words like "need" and "deserve."

When clients have financial difficulty, the rehabilitation veterinarian and technician/nurse need to discuss privately whether pursuing advanced diagnostics will change the course of treatment. Does the client have the resources to pursue corrective surgery? What about an orthosis or prosthesis? The client should never be shamed into feeling that they are inadequately caring for their pet. The technician/nurse can be an advocate for the client when they are too embarrassed to speak up about costlier treatments. If the client cannot afford a full rehabilitation plan, then an alternative plan can be put into place where the client can be taught some rehabilitation techniques to be performed at home with an agreement for in clinic recheck examinations as often as possible. When talking to owners about exercise, comprehensive instructions need to be given about limitations, frequency of walks and therapeutic exercise, and consistency. Dissuade owners from overexercising their dogs at weekends. Types of exercises are all explained in detail in Chapter 6.

## Supporting the Patient

The quality of pain management in practices seems to be directly related to veterinary technicians and nurses. This includes the physical rehabilitation veterinary technician/nurse. The role of advocate for a non-verbal patient can be daunting. Veterinary technicians and nurses are in the unique position of being responsible for most of the patient care and and its quality without the freedom to prescribe or initiate therapy (Shaffran, 2008). Knowledge of the appropriate techniques, modalities, therapies, and equipment prescribed by the rehabilitation veterinarian is essential for good communication between veterinarians and veterinary technicians/nurses. The skilled technician is a source of vital information for the rehabilitation veterinarian every day in practice. Technicians use critical thinking, observation, and interpretive skills to make important recommendations. Discussion of each case directly with the veterinarian might include the technician's particular concerns about a patient. Based on his or her interaction with patients, the technician may offer suggestions for adjustments, changes, or additions to the program. Giving technicians a voice in the rehabilitation process creates a truly positive team environment in which their thoughts and skills are valued. Daily medical rounds are important to the rehabilitation team, allowing communication encompassing patient advocacy. Patients ultimately receive better care when a technician can advocate, and technicians are satisfied knowing that they are doing everything they can to ensure the well-being of patients in their charge.

## Improving Quality of Life

Quality of life (QoL) is best approached by first deciding what is important for the animal, and second by working out what can be assessed for use in decision-making. This requires a combination of assessing those qualities of life from the animal's point of view and the assessment from the observer's point of view. Animal QoL includes the feelings of the animal, which can be broadly classified as pleasant or unpleasant (Kerrigan, 2014a). Using the analogy of balance scales, QoL can be improved by increasing the pleasant feelings and decreasing the unpleasant ones. The veterinary team and client should partner to consider the current QoL of the pet and identify ways in which this can be maintained as the pet progresses through their senior years.

Appropriate analgesics should be used for conditions that can affect QoL. An example is for the most common musculoskeletal condition affecting dogs and cats, osteoarthritis. Osteoarthritis may affect up to 20% of dogs over 1 year of age, and nearly 50% of musculoskeletal disorders identified in a 10-year span in 16 veterinary hospitals resulted from joint disease (Canapp, 2013). In 2002, Hardie *et al.* examined skeletal radiographs of 100 cats over 12 years of age and found that over 90% of cats had radiographic evidence of degenerative joint disease. In addition to the use of analgesics and supplements (Cotman *et al.*, 2002; Fritsch *et al.*, 2010), environmental management and modification can provide some easy ways for clients to enhance the QoL of the arthritic pet's everyday world. Modifications of sleeping surface, eating bowls, and home flooring can help pets with compromised mobility (see Chapters 10 and 12). There are many anxiety and depression treatments that can help a pet to be more calm (allowing the client to be less worried). These treatments range from pheromone collars to supplements, such as milk proteins, through to medications (Roush *et al.*, 2010).

Nutritional counseling should be available to the client. Veterinary technician/nurses often play a role in the nutritional evaluation process and it is essential to standardize the procedure. A good starting point to ensure consistency among team members and to focus on evidence-based research is the global nutrition guidelines of the World Small Animal Veterinary Association (WSAVA, 2011) or the nutritional assessment guidelines of the American Animal Hospital Association (AAHA, 2010).

It is important to advise clients to create frequent moments of enjoyment for their mobility-challenged companion through environmental enrichment. An aging pet, for example, may be less mentally alert and responsive, which can be mistaken by owners as "old age" stubbornness or a lack of interest in playtime activities (Kerrigan, 2014a). Environmental enrichment should focus on positive interactions such as petting or massage, as well as new and varied opportunities for exploration, including different walks/surroundings, find-and-seek games and other stimulating ways to obtain food and treats. Food toys which require pushing, lifting, pawing, or rolling to release food can help aging pets to remain active and alert (Landsberg *et al.*, 2012).

## Mobility

As mentioned above, home modifications can be made to support the physically challenged patient and to enable an improvement in mobility. Mobility issues will always affect QoL to some extent, but with the right aids a pet can have a good life. The needs of both the human and the pet need to be balanced. For details about mobility issues and aids, see Chapter 10 on assistive devices. Mobility changes in late life may include frailty syndrome. This is a term from human medicine (Cesari *et al.*, 2016; Ekdahl *et al.*, 2016), which describes a decline in the body's functional reserve, lower energy metabolism, smaller muscle cells, and altered hormonal and inflammatory functions. Some of these signs have been identified in dogs (DeLorey *et al.*, 2012; Wallis *et al.*, 2016) (Figure 7.4) and the authors have seen them in cats. Frailty leads to increased susceptibility to disease and functional dependency which can be a huge strain on both human and pet (see Chapter 12).

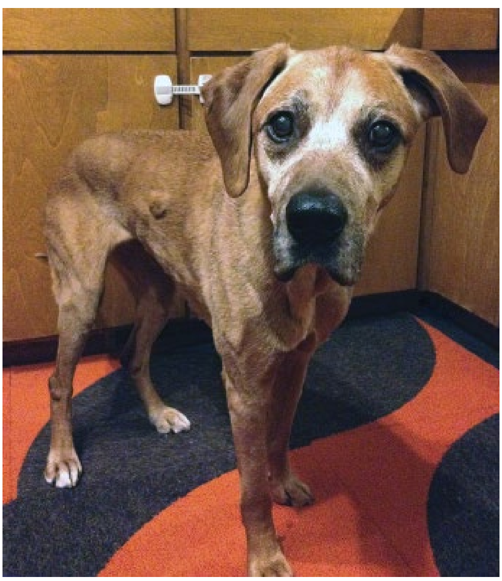

Figure 7.4 Elderly dog with frailty syndrome. Muscle loss is seen by evident bony prominences.

## Behavioral Changes

Behavioral changes can affect and even sever the human–animal bond. These changes can occur at any stage of life. An example of a disease causing behavioral changes is cognitive dysfunction (Rème *et al.*, 2008). Clinically, cognitive dysfunction may result in various behavioral signs, including disorientation; forgetting of previously learned behaviors, such as house training; alterations in the manner in which the pet interacts with people or other pets; onset of new fears and anxiety; decreased recognition of people, places, or pets; and other signs of deteriorating memory and learning ability (Landsberg and Araujo, 2005; Araujo *et al.*, 2008). Behavioral signs related to anxiety, vocalization, night waking, soiling in cats, and aggression in dogs are more often spontaneously reported to veterinarians, which is likely related to the impact of these behaviors on the owner (Landsberg *et al.*, 2012). Companion animal relinquishment has been defined as when an owner voluntarily gives up ownership or possession of their pet. This includes surrender, euthanasia, and abandonment (Coe *et al.*, 2014). According to the National Council on Pet Population Study and Policy, shelters in the United States euthanize 72% of relinquished cats, many because of house-soiling behavior (Carney *et al.*, 2014). Changes in behavior can occur with additions to household, loss of a special human, moving home location, and other stressors. One such stressor may be the home exercise plan prescribed. When teaching home exercises, using positive reinforcement and food motivation is the best way to prevent or reduce stress between human and pet (see Chapter 9). Be sure to check in with the client about home exercises, having the client demonstrate again at the clinic and assessing the pet's behavior when doing the home exercises in clinic. The rehabilitation team needs to rule out pain as a cause of poor compliance and to revise the plan as needed. For a patient coming to the end of life, any small change in QoL can make a huge difference. Consult a veterinary behavior specialist if needed. Remember that behavior changes may be part of more global nervous system signs in the cause of cerebral disease or neoplasia.

## Veterinary Home Hospice Care

The American Veterinary Medical Association (2014) stated that hospice care offered within the context of veterinary practice or the home environment, and consistent with veterinary practice legislation, gives clients time to make decisions regarding a companion animal with a terminal illness or condition and to prepare for the impending death of that animal (Kerrigan, 2014b). If home care is going to be utilized, then consent must be gained from the client. Many mobile veterinarians will visit the client and patient in the home to assess the current situation. The client's record should be updated to state that the patient is terminal and/or incapacitated and is being cared for at home with appropriate analgesia and limited supportive care at the bequest of the family for hospice care and euthanasia once QoL becomes inappropriate (Villalobos, 2009).

For both humans and animals, the most common condition necessitating hospice care is cancer. One key difference between humans and animals with regard to hospice care options is the availability of euthanasia for animals. A pet owner's goals and priorities may vary and transform as morbidity progresses, therefore, they should be made aware that if they commit to a home-care hospice program, the situation will be closely monitored and if at any point management becomes difficult the situation will be reviewed. A supportive, coaching approach from the veterinary team is essential to help clients address the needs of their pet at home. The veterinarian will provide the treatment care plan and the veterinary technician/nurse can be supportive, providing educational materials and demonstrations. The level of care needed for the pet should be thoroughly discussed so that it may be ascertained how much the pet owner can contribute to the level of care required (Hancock *et al*., 2004). This will dictate how much external care is required and if, in fact, a home-care program is a viable option for this pet and owner (Kerrigan, 2014b).

Incontinence is an issue that should be discussed with the family members as many end-of-life patients will develop urinary and/or fecal incontinence. Owners must be educated regarding the prevention of urine and fecal scald along with the basics of good hygiene practices. The veterinary nurse can offer advice regarding the wearing of personal protective equipment and appropriate disposal of soiled materials along with safe disposal of any sharps used. These are important considerations about which pet owners must be thoroughly informed prior to committing to a home-care hospice program. More about continence issues and home care can be found in Chapter 11. Ideally, a social location within the home should be selected as the pet's designated area. This will enable the pet to be part of the normal family activities, especially if this is what they had been used to doing. All areas in which the pet will reside must have access to drinking water in order that the animal does not have to physically move to another location to get a drink; such a strategy will help to minimize the risk of dehydration (Kerrigan, 2014b).

## Palliative Care – Fluids at Home, Injectable Medications as well as Oral

Provision of home hospice care will require the owner, and possibly other family members, to provide medication and care techniques detailed in the care plan. The veterinary technician or nurse could demonstrate how to administer a subcutaneous injection; the pet owner would then demonstrate the technique back to the veterinary nurse in order that their level of competency may be assessed. Written instructions should also be provided to the client for clarification and review. By educating clients to provide certain types of care in their home, it helps them to gain confidence and feel they have a sense of control. This improves coping strategies in stressful circumstances. Some specific veterinary practices allow the veterinary technician/nurse to visit the client's home, under the veterinarian's direction, to assess QoL, perform appropriate nursing interventions, and provide encouragement and support during this difficult period.

A practice vehicle will be required to make home visits. In addition to the initial financial outlay of this, running costs, fuel, and insurance will all need to be factored into the overall cost of the program (Kerrigan, 2014b).

At some point, it may become necessary for temporary hospitalization of the patient. Hospice pets often require special boarding care or daycare with their veterinary team for supervision, wound care, hygiene, hand feeding, and even placement of esophagostomy tube to help the patient get over a period of not eating (Villalobos, 2004). This care can and should be willingly provided at the primary care veterinary hospital. This service may be the key to sustaining a hospice pet for working owners (respite care). Be sure to arrange convenient drop-off and pickup times that fit the client's working and travel schedule.

## Other Medication Considerations

For pets that are difficult to medicate, oral medications may be compounded into a more palatable medication or administered subcutaneously if owners are comfortable and competent with the procedure. It may also be possible to reconstitute medication to a transdermal or transmucosal formulation (McVety, 2012). Not all transdermal medications are absorbed systemically to therapeutic levels. Your veterinarian will be aware of those that are scientifically verified. What must be of paramount concern here is that the patient's level of analgesia is not severely compromised because it is being cared for in the home environment. A pain scale should be taught to the owners so that daily pain score can be recorded and assessed. It is necessary to have a pain scale that can encompass both aspects of acute and chronic pain. This ensures that the client working with veterinary personnel can help the pet owner to anticipate, prevent, locate, and relieve pain in the in-home hospice patient. A review of pain management can be found in Chapter 3.

## When It Is Time to Say Goodbye

Veterinarians are the individuals trained to recognize when QoL has declined and suffering has become unacceptable. When discussing end of life, the team has to assess two things – the patient's QoL and the client's feelings. As part of a patient's therapy team the technician/nurse has an important role.

There is a QoL scale developed by Dr. Alice Villalobos called the HHHHHMM scale. Each of the categories is rated on a 1–10 scale (10 is best):

- Hurt
- Hunger
- Hydration
- Hygiene
- Happiness
- Mobility
- More good days than bad.

A score of 35 or more out of 70 reflects an acceptable QoL. This scale can help a client to understand their pet's needs but other factors may play a part in the client's decision for their pet. Regarding pain level, you and your veterinarian need to educate the client. Home behaviors will be different to those in clinic and need to be recorded and taken into account. This scale can be found at Dr. Villalobos's webpage (http://pawspice.com/).

How you navigate the discussion with the owner is not difficult but the topic needs to be approached with caution. Questions such as "How have things changed for your pet?"

and "Is he still excited to see you when you get home?" can open up conversation. Let them talk to you – they may tell you a lot more than they tell the veterinarian. Alert your veterinarian in your regular communications at patient rounds. Recommend to the client that they make an appointment to "check in" with the veterinarian. Take time, even if you are running late.

#### Find a Way to End on a Good Note

Work to understand the client's decision. Even if you are not ready to say goodbye, you too have a strong bond with the pet. Remember the load on a primary caregiver can be huge and home behaviors and factors may be worse than described. Use words carefully: "I am sorry you and Spot are going through this." Beware of the question "What would you do?" Answer taking into account information you have so far and using your knowledge of this client's personality. Be sure to stress that the decision is a very individual one.

#### After Loss

Send a card to express your sentiments but do not imply your grief is greater! Remember that client beliefs may differ from yours. Grief counseling is available; it is a difficult thing to suggest.

Keep a list of bereaved clients and contact them when you see a rescue dog or cat. Even if they are not ready, they will be glad that you thought of them.

## Conclusion

Communication is the key to supporting both client and patient through the rehabilitation process, whether it is a relatively short-term process, with a return to good function for the patient or a long-term period of care that is ongoing until end of life. The veterinary technician/nurse plays a huge role in communicating the client's needs and wishes to the rehabilitation veterinarian. You will help to educate the clients in order to provide the best care that they can give to their beloved pet. Rehabilitation is a large commitment for a client and pet as well as for the rehabilitation team. Regular communication, counseling about expected setbacks, and providing short- and long-term goals can help to keep everyone in the clinic on message and the client aware of a cohesive, committed approach to care.

## References

AAHA (American Animal Hospital Association) (2003) *The Path to High Quality Care: Practical Tips for Improving Compliance*. https://openlibrary.org/works/OL5754480W/The_Path_to_High_Quality_Care (accessed June 2017).

AAHA (2010) AAHA nutritional assessment guidelines for dogs and cats. *J Am Anim Hosp Assoc* 46:285–296. https://www.aaha.org/public_documents/professional/guidelines/nutritionalassessmentguidelines.pdf

American Veterinary Medical Association (2014) *Guidelines for Veterinary Hospice Care*. https://www.avma.org/KB/Policies/Pages/Guidelines-for-Veterinary-Hospice-Care.aspx (accessed February 3, 2016).

Araujo JA, Landsberg GM, Milgram NW, and Miolo A (2008) Improvement of short-term memory performance in aged beagles by a nutraceutical supplement containing phosphatidylserine, *Ginkgo biloba*, vitamin E, and pyridoxine. *Can Vet J* 49(4):379–385.

Balducci L (2007) Caring for geriatric cancer patients: What is needed aside from medical care? In *Canine and Feline Geriatric Oncology – Honoring the Human–Animal Bond* (eds. A Villalobos and L Kaplan). Blackwell Publishing, Ames, IA.

Bellows J, Colitz CM, Daristotle L, *et al.* (2015) Common physical and functional changes associated with aging in dogs. *J Am Vet Med Assoc* 246(1):67–75.

Bowlby J (1969) *Attachment and Loss*, Vol. 1: *Attachment*. Basic Books, New York, 448 pp.

Bretherton I (1992) The origins of attachment theory: John Bowlby and Mary Ainsworth. *Dev Psychol* 28:759–775.

Canapp D (2013) Canine osteoarthritis. *Clinician's Brief* August: 21–23.

Carney HC, Sadek TP, Curtis TM, *et al.* (2014) AAFP and ISFM guidelines for diagnosing and solving house-soiling behavior in cats. *J Feline Med Surg* 16:579–598.

Carter CS (1998) Neuroendocrine perspectives on social attachment and love. *Psychoneuroendocrinology* 23:779–818.

Cesari M, Prince M, Thiyagarajan JA, *et al.* (2016) Frailty: An emerging public health priority. *J Am Med Dir Assoc* 17(3):188–192.

Coe JB, Young I, Lambert K, *et al.* (2014) A scoping review of published research on the relinquishment of companion animals. *J Appl Anim Welf Sci* 17(3):253–273.

Cotman CW, Head E, Muggenburg BA, *et al.* (2002) Brain aging in the canine: a diet enriched in antioxidants reduces cognitive dysfunction. *Neurobiol Aging* 23(5):809–818.

DeLorey DS, Clifford PS, Mittelstadt S, *et al.* (2012) The effect of aging on adrenergic and nonadrenergic receptor expression and responsiveness in canine skeletal muscle. *J Appl Physiol* 112:841–848.

Downing R (2011a) Pain management for veterinary palliative care and hospice patients. *Vet Clin Small Anim* 41:531–550.

Downing R (2011b) The role of physical medicine and rehabilitation for patients in palliative and hospice care. *Vet Clin Small Anim* 41:591–608.

Ekdahl AW, Alwin J, Eckerblad J, *et al.* (2016) Long-term evaluation of the ambulatory geriatric assessment: A frailty intervention trial (AGe-FIT): Clinical outcomes and total costs after 36 months. *J Am Med Dir Assoc* 17(3)263–268.

Fallani G, Prato-Previde E, and Valsecchi P (2006) Do disrupted early attachments affect the relationship between guide dogs and blind owners? *Appl Anim Behav Sci* 100:241–257.

Fritsch DA, Allen TA, and Dodd CE (2010) A multicentre study of the effect of dietary supplementation with fish oil omega-3 fatty acids on carprofen dosage in dogs with osteoarthritis. *J Am Vet Med Assoc* 236(5):535–539.

Hancock CG, McMillan FD, and Ellenbogen TR (2004) Owner services and hospice care. In *Geriatrics and Gerontology of the Dog and Cat* (ed. J Hoskins). Saunders, St. Louis, MO.

Hardie EM, Roe SC, and Martin FR (2002) Radiographic evidence of degenerative joint disease in geriatric cats: 100 cases (1994–1997). *J Am Vet Med Assoc* 220:628–632.

Jurek C and McCauley L (2011) Physical rehabilitation & senior pets. *NAVC Clinician's Brief* December:59–65.

Kerrigan L (2013) Veterinary palliative and hospice care – making the transition from 'cure' to 'care'. *Vet Nurse* 4(6):316–321.

Kerrigan L (2014a) How to preserve quality of life in the ageing canine: in-home modification and nutritional supplements. *Vet Nurse* 5(6):338–343.

Kerrigan L (2014b) In-home hospice provision – a viable option for veterinary palliative care? *Vet Nurse* 5(3):146–151.

Kubinyi E, Topál J, Miklósi Á, and Csányi V (2003) Dogs (*Canis familiaris*) learn from their owners via observation in a manipulation task. *J Comp Psychol* 117:156–165.

Landsberg G and Araujo JA (2005) Behavior problems in geriatric pets. *Vet Clin Small Anim* 35:675–698.

Landsberg GM, Nichol J, and Araujo JA (2012) Cognitive dysfunction syndrome: a disease of canine and feline brain aging. *Vet Clin North Am Small Anim Pract* 42(4):749–768.

McVety D (2012) Veterinary hospice: Medicate, mediate, mitigate. *NAVC Clinician's Brief* February:33–36.

Rehn T, McGowan RT, and Keeling LJ (2013) Evaluating the strange situation procedure (SSP) to assess the bond between dogs and humans. *PLoS One* 8(2):e56938.

Rème CA, Dramard V, Kern L, *et al.* (2008) Effect of S-adenosylmethionine tablets on the reduction of age-related mental decline in dogs: a double-blinded, placebo-controlled trial. *Vet Ther* 9(2):69–82.

Roush JK, Cross AR, and Renberg WC (2010) Evaluation of the effects of dietary supplementation with fish oil omega-3 fatty acids on weight bearing in dogs with osteoarthritis. *J Am Vet Med Assoc* 236(1):67–73.

Serpell JA (2003) Anthropomorphism and anthropomorphic selection: Beyond the "cute response". *Soc Anim* 11:83–100.

Shaffran N (2008) Pain management: the veterinary technician's perspective. *Vet Clin Small Anim* 38:1415–1428.

Sharp B (2008) Physiotherapy in small animal practice. *In Practice* 30:190–199.

Smith B (2012) The 'pet effect' – health related aspects of companion animal ownership. *Aust Fam Physician* 41(6):439–442.

Topál J, Miklósi Á, Csányi V, and Dóka A (1998) Attachment behavior in dogs (*Canis familiaris*): A new application of Ainsworth's (1969) Strange Situation Test. *J Comp Psychol* 112:219–229.

Villalobos AE (2004) *Canine and Feline Geriatric Oncology: Honoring the Human–Animal Bond*. Blackwell Publishing, Ames, IA.

Villalobos AE (2009) Hospice: A way to care for terminal pets. *NAVC Clinician's Brief* May:39–45.

Vitale Shreve KR and Udell MA (2015) What's inside your cat's head? A review of cat (*Felis silvestris catus*) cognition research past, present and future. *Anim Cogn* 18(6):1195–1206.

Wallis LJ, Virányi Z, Müller CA, *et al.* (2016) Aging effects on discrimination learning, logical reasoning and memory in pet dogs. *AGE* 38:1–6.

Wright B (2013) Pain management in hospice and palliative care. *Proceedings for the International Association of Animal Hospice and Palliative Care*, Third Annual Animal Hospice Meeting, Denver, CO. October 17–20, 2013.

WSAVA (World Small Animal Veterinary Association) (2011) *Global Nutrition Guidelines*. http://www.wsava.org/guidelines/global-nutrition-guidelines

# 8

# Nutritional Counseling

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CHAPTER MENU



## Introduction

Nutrition plays a vital role in the prevention and management of many conditions seen by veterinary health care team members. This can especially be seen in the rehabilitation of certain disease conditions and should be considered an integral component of the physical rehabilitation protocol. This chapter will look at how nutrition supports the goals of veterinary rehabilitation as well as the technician's role in counseling clients on the importance of nutrition.

## Nutritional Assessment

Proper nutrition is a critical component for maintaining the health of pets. Every patient, healthy or ill, that enters the veterinary hospital should have an evaluation of their nutritional status and health care team members should make a nutritional recommendation based on this evaluation. The goal of patient assessment is to establish the key nutritional factors and their target levels in light of the patient's physiologic or disease condition. This assessment is vital and is, in fact, the fifth vital assessment to be performed by the health care team.

The nutritional assessment considers several factors, including the animal, the diet, feeding management, and environmental factors. The nutritional assessment is an iterative process in which each factor affecting the animal's nutritional status is assessed and reassessed as often as required (Thatcher *et al.*, 2010).

Although the impact of nutrition on health is a complicated area of study, the American

*Physical Rehabilitation for Veterinary Technicians and Nurses*, First Edition.
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Animal Hospital Association (AAHA) and the World Small Animal Veterinary Association (WSAVA) nutritional guidelines can be abridged into three steps:

1) Incorporate a nutritional assessment and specific dietary recommendation in the physical exam for every pet, every visit.
2) Perform a screening evaluation (nutritional history/ activity level, body weight and body/muscle condition score) – every pet, every visit.
3) Perform an extended evaluation for a patient with abnormal physical exam findings or nutritional risk factors such as lifestage considerations, abnormal body condition score or muscle condition score, poor skin or hair coat, systemic or dental disease, diet history of snacks or table food that is greater than 10% of total calories, unconventional diet, gastrointestinal upset, or inadequate or inappropriate housing.

## Obesity

Pet obesity has reached epidemic proportions in the United States and other industrialized countries. It is estimated that 35% of adult pets and 50% of pets over age 7 are overweight or obese (German *et al*., 2006). Obesity can be defined as an increase in fat tissue mass sufficient to contribute to disease. Dogs and cats weighing 10–19% more than the optimal weight for their breed are considered overweight; those weighing 20% or more above the optimum weight are considered obese. Obesity has been associated with a number of diseases as well as with a reduced lifespan. A combination of excessive caloric intake, decreased physical activity, and genetic susceptibility are associated with most cases of obesity and the primary treatment for obesity is reduced caloric intake and increased physical activity. Obesity is one of the leading preventable causes of illness/death and with the dramatic rise in pet obesity over the past several decades, weight management and obesity prevention should be among the top health issues health care team members discuss with every client.

### Causes of Obesity

Obesity is an imbalance of energy intake and energy expenditure. It is a simple equation – too much in, too little out! There are a number of risk factors that affect energy balance. Indoor pets in North America are typically neutered. There are many positive health benefits associated with neutering, however it is important that metabolic impacts of neutering are addressed as well. Studies have demonstrated that neutering may result in decreased metabolic rate and increased food intake, and if energy intake is not adjusted, body weight, body condition score (BCS), and amount of body fat will increase resulting in an overweight or obese pet. Other recognized risk factors for obesity include breed, age, decreased physical activity, and type of food and feeding method (Burkholder and Toll, 2000; Rosenthal, 2007).

### Health Risks Associated with Obesity

There are many health conditions associated with obesity in pets, including arthritis, diabetes mellitus, cancer, skin diseases, lower urinary tract problems, hepatic lipidosis, and heart disease. Obese pets are also more difficult to manage in terms of sample collection (blood, urine) and catheter placement and may be more prone to treatment complications, including difficulty intubating, respiratory distress, and slower recovery time and delayed wound healing. It is widely believed that obesity affects quality of life and leads to reduced life expectancy. The dramatic impact of excess body weight in dogs and cats has been demonstrated. In cats, it is estimated that 31% of diabetes mellitus and 34% of lameness cases could be eliminated if cats were at optimum body weight. In dogs, lifespan was increased by nearly 2 years in dogs that were maintained at an optimal body condition (Burkholder and Toll, 2000). It is important to recognize and to communicate to our clients that fat tissue is not inert – obesity is not an aesthetic condition that only affects our pet's ability to interact with us on a physical activity level. Fat tissue is metabolically active and, in fact, is the largest endocrine organ in the body and has an unlimited growth potential. Fat tissue is an active producer of hormones and inflammatory cytokines and the chronic low-grade inflammation secondary to obesity contributes to obesity-related diseases (Burkholder and Toll, 2000; Laflamme, 2006).

## Evaluating Weight and Nutrition

Most pet owners do not recognize or admit that their pet is overweight. The entire health care team needs to commit to understanding and communicating the role of weight management in pet health and disease prevention. In particular, the veterinary technician is the primary source for client education – the interface between the client, the doctor and the rest of the hospital team – and is the key advocate for the patient.

Again, every patient needs an assessment to establish nutritional needs and feeding goals. These goals will vary depending on the pet's physiology, obesity risk factors, and current health status. Designing and implementing a weight management protocol supports the team, the client and, most importantly, the patient (Rosenthal, 2007).

A comprehensive history including a detailed nutritional profile and a thorough physical examination, including a complete blood count, serum chemistry, and urinalysis, are the first steps in patient evaluation. Signalment data should include species, breed, age, gender, neuter status, weight, activity level, and environment. The nutritional history should determine the type of food (all food) fed, the feeding method (how much, how often), who is responsible for feeding the pet, and any other sources of energy intake (no matter how small or seemingly insignificant) (Box 8.1).

Obtaining a complete nutritional history supports consistency and accuracy of patient information, helps to pinpoint potential barriers to client compliance, guides client discussion, and supports the optimal weight management program for the pet.

Be sure to weigh the pet and obtain a BCS at every visit and record the information in the patient's medical record. It is helpful to use the same scale and chart the findings for the client. BCS is important to assess a patient's fat stores and muscle mass. A healthy and successful weight management program results in loss of fat tissue while maintaining lean body mass and consistent and accurate assessment of weight and BCS is an important tool to track

**Box 8.1 Questions for nutritional assessment**

The following questions should be part of every nutritional assessment:

- What brand of food do you feed your pet? (try to get specific name)
- Tell me what your pet eats in a day
- Do you feed moist or dry or both?
- How do you feed your pet – feeding method (how much, how often)?
- Does your pet receive any snacks or treats of any kind? If so, what and how often?
- Do you give your pet any supplements?
- Is your pet on any medications, including chewable medications? If so, obtain name and dosage
- What type of chew toys does your pet play with?
- Do you feed your pet any foods or treats not specifically designated for pets (such as human foods)? If so, what and how often?
- Does your pet have ANY access to other sources of food (neighbor, trash, family member, etc.)?

progress. The use of body condition charts and breed charts are helpful tools in discussing the importance of weight management with clients and will help them visualize what an optimal weight would look like on their pet.

Although subjective, BCS is a tool to evaluate body fat. The goal for most patients is a body condition of 2.5–3.0 out of 5 or 4–5 out of 9, which is considered to be ideal (Figures 8.1 and 8.2).

An ideal BCS is believed to help decrease health problems, including musculoskeletal conditions. The evaluation of muscle mass independent from body fat content assessed by BCS, is known as muscle condition scoring. Muscle condition scoring includes visual examination and palpation over temporal bones, scapulae, lumbar vertebrae, and pelvic bones. The loss of muscle unfavorably affects the strength, immune function, and wound healing of pets. After all information is collected and analyzed in the assessment phase a treatment plan is implemented. This is followed by repeated assessment and adjustment of the plan (Figures 8.3 and 8.4).

A newer tool in obtaining an accurate fat percentage on the pet's body is Body Fat Indexing (BFI) – especially for overweight and obese pets (Ahima, 2006; Wortinger and Burns, 2015a). This method of obtaining a more accurate body fat percentage has been utilized and confirmed by veterinary nutritionists and provides a better method for pinpointing the amount of fat on a specific pet, thus aiding in accurately calculating an amount to feed for an overweight pet, better identifying the pets' ideal body weight, insuring proper and safe weight loss, and increasing the overall success in the weight management program.

## Weight Management Program

As with many aspects of health care, designing a successful weight management program is not a "one program fits all" for our patients. The components of a successful weight management program include consistent and accurate weight measurement/patient monitoring, effective client communication, identification of compliance gaps and utilization of tools to reinforce compliance, client and patient support, and program restructure as needed.

Setting a goal for weight loss and calculating the appropriate energy intake starts with determination of the pet's ideal body weight. Ideal body weight is a starting goal that is adjusted for appropriate body condition as the pet loses weight. It is important to determine the number of daily calories that will result in weight loss while providing adequate protein, vitamins, and minerals to meet the pet's daily energy requirement (DER). The DER reflects the pet's activity level and is a calculation based on the pet's resting energy requirement (RER).

There are a couple of basic formulas that all technicians should memorize or have on laminated note cards in every exam room (along with a calculator)! The most accurate formula to determine the RER for a cat or a dog is:

$$\text{RER kcal/day} = 70(\text{ideal body weight in kg})^{0.75}$$

RER is determined initially, followed by the DER. DER may be calculated by multiplying RER by "standard" factors related to energy needs. The calculations used to determine energy needs for obese prone pets or for pets needing to lose weight are (Rosenthal, 2007; Wortinger and Burns, 2015a):

- Obese prone dogs DER = 1.4 × RER
- Weight loss/dogs DER = 1.0 × RER
- Obese prone cats DER = 1.0 × RER
- Weight loss/cats DER = 0.8 × RER

Gathering the above information is crucial and takes only a few minutes. This information provides the foundation for developing a weight loss program that includes:

1) Target weight or weight loss goal
2) Maximum daily caloric intake
3) Specific food, amount of food and method of feeding.


Figure 8.1 WSAVA body condition score for cats. *Source:* http://www.wsava.org/sites/default/files/Body Condition Scoring for Cats.pdf


Figure 8.2 WSAVA body condition score for dogs. *Source:* http://www.wsava.org/sites/default/files/Body%20condition%20score%20chart%20dogs.pdf

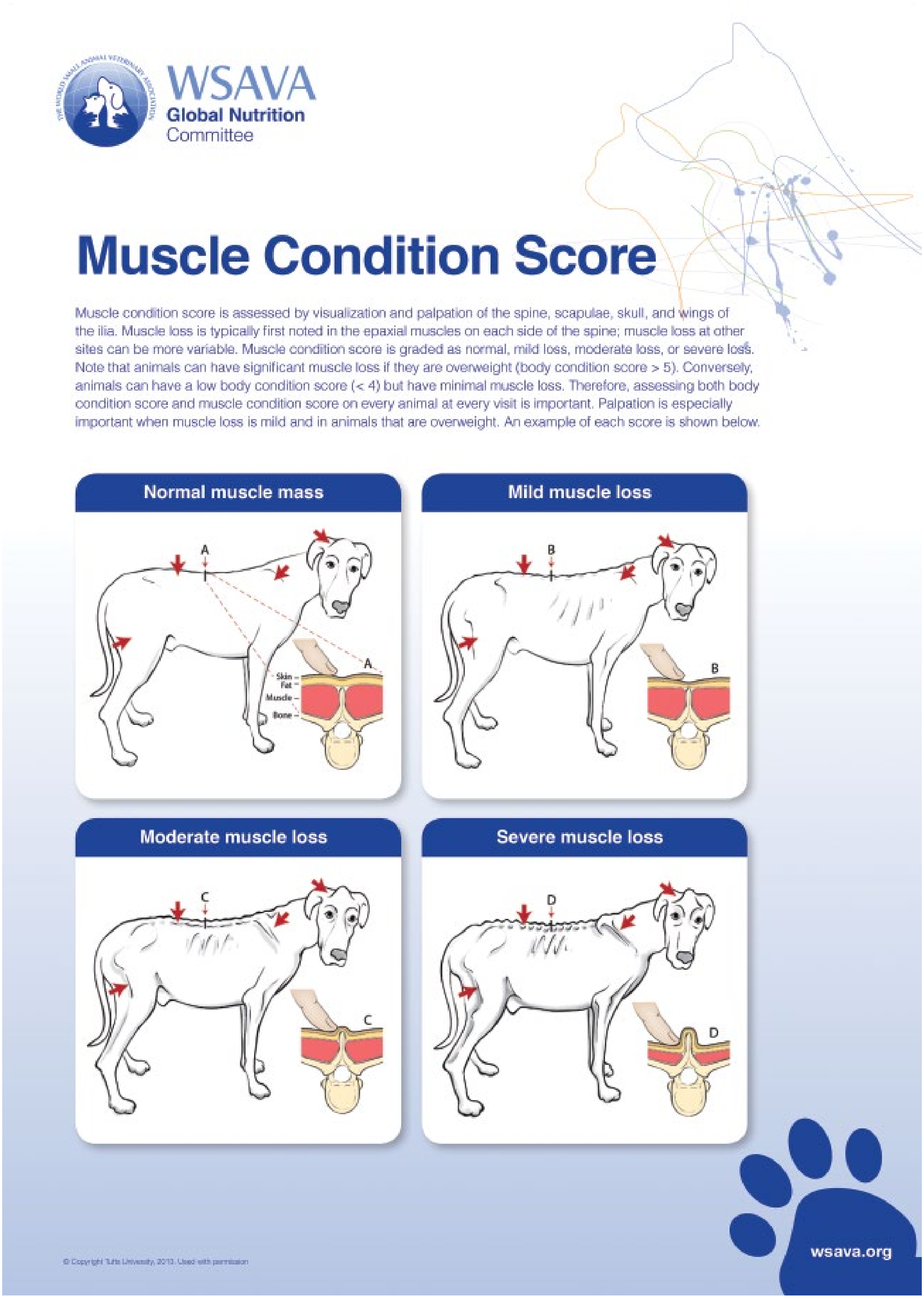


**Figure 8.3** WSAVA muscle condition score for dogs. *Source:* http://www.wsava.org/sites/default/files/Muscle%20condition%20score%20chart%202013.pdf

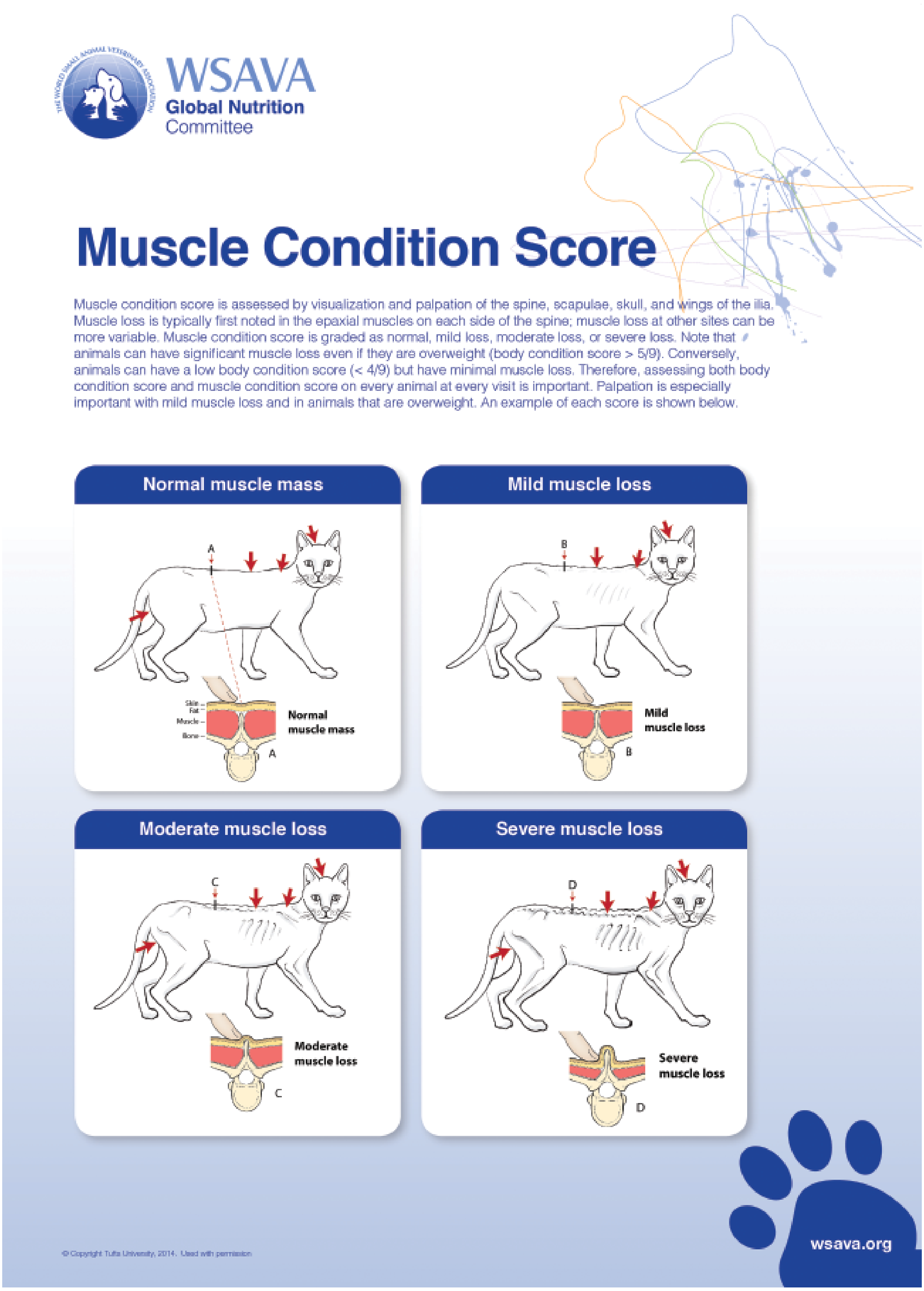


**Figure 8.4** WSAVA muscle condition score for cats. *Source:* http://www.wsava.org/sites/default/files/Muscle%20condition%20score%20chart-Cats.pdf

**Box 8.2 Specific recommendations that support a successful weight loss program**

- Feeding consistency (same time, same amount, same place, same dish, etc.) including feeding the pet-designated dish only
- Ensure the use of an 8 ounce (240 mL) measuring cup
- Recommend the appropriate weight loss food and calculate the initial amount to feed
- Discuss the importance of total energy intake (do not feed anything other than the recommended food at the designated amount)
- Make appropriate recommendations for treats and adjust the caloric intake of the base food accordingly
- Encourage client's to feed their pet's separately if possible
- Recommend appropriate exercise for the pet
- Offer suggestions on ways other than food to reward or bond with their pet
- Evaluate, adjust, communicate, and encourage on a consistent basis

The program should also include specific protocols for monitoring the pet's weight (schedule these before the client leaves and send reminder cards/e-mails/texts). Also prior to leaving, the health care team should adjust the pet's energy intake and exercise guidelines/suggestions accordingly (Box 8.2).

Successful weight management begins with recognition of the importance of weight control in our pets. It is essential that the health care team, specifically the veterinary technician, communicate the serious effects that even a few excess pounds can have on the health and longevity of their pet's lives. Weight management should be a cornerstone wellness program in every clinic and the veterinary technician the champion of the program and advocate for the patient.

## Osteoarthritis

As in humans, osteoarthritis is the most common form of arthritis recognized in all veterinary species. A slowly progressive condition, osteoarthritis is characterized by two main pathologic processes: (i) degeneration of articular cartilage with a loss of both proteoglycan and collagen and (ii) proliferation of new bone. Furthermore, there is a variable, low-grade inflammatory response within the synovial membrane (Anandacoomarasamy *et al.*, 2008). Current estimates of the prevalence of arthritis in senior and geriatric dogs range from 20% to 25%. The prevalence of osteoarthritis in adult cats is 33%, with the prevalence in senior cats rising to 90% (Lascelles and Robertson, 2010). The objectives of treatment for osteoarthritis are multifaceted: reduce pain and discomfort, decrease clinical signs, slow the progression of the disease, promote the repair of damaged tissue, and improve the quality of life. A multimodal approach to osteoarthritis provides the best results for dogs with chronic pain due to osteoarthritis and include a combination of anti-inflammatory and analgesic medications, disease-modifying osteoarthritis agents (DMOAs), nutraceuticals, weight reduction, exercise programs, physical therapy, and therapeutic foods. Applying an individualized combination of these management options to each patient will enhance quality of life, which is the ultimate goal of therapy.

### Weight Reduction

We have discussed the fact that obesity is epidemic in companion animals. Numerous studies indicate that 25–40% of adult dogs are overweight or obese. Similar to the Centers for Disease Control (CDC) estimate that approximately 33% of all human adults suffer from arthritis, an estimated 20% of the adult canine population suffers from osteoarthritis. One long-term study has documented that the prevalence of osteoarthritis is greater in overweight/obese dogs than in ideal weight dogs (83% vs. 50%) (Kealy *et al*., 2000; Burns, 2011). Consequently, it is reasonable to assume a significant portion of arthritic dogs will be overweight/obese and vice versa. Managing these concurrent conditions presents many challenges.

Before a disease can be treated it must first be diagnosed. As disease entities, osteoarthritis and overweight/obesity present diagnostic challenges for very different reasons. Clinical signs of osteoarthritis are often not obvious on examination, particularly early in the disease process. Signs of overweight/obesity may be readily apparent, but often they are overlooked or dismissed as inconsequential. Diagnosis of osteoarthritis requires a combination of history, physical examination findings, and radiographic evidence of degenerative joint disease. Although this may seem straight forward, historical clues crucial to diagnosis may not be readily apparent on routine veterinary examination. Owners often believe many signs of osteoarthritis are a part of "normal" aging and consequently fail to report them unless prompted by the health care team.

Clinical signs of advanced arthritis include difficulty rising from rest, stiffness, or lameness. A thorough, disease-specific history should be taken and may reveal evidence of subtle changes early in the course of osteoarthritis, such as reluctance to walk, run, climb stairs, jump, or play. Signs may be as discreet as lagging behind on walks. Owners are often unaware of the correlation between behavior changes and arthritis. Yelping or whimpering and even personality changes (i.e., withdrawal, aggressive behavior) may be indicative of the chronic pain of osteoarthritis. It is recommended to use an owner questionnaire with every potential osteoarthritis patient to assist with early detection of osteoarthritis.

Recognizing signs of osteoarthritis in cats is much more difficult. Cats often suffer in silence and the veterinary health care team must rely upon the owner's evaluation and a thorough history to discover potential signs and symptoms of osteoarthritis in cats. The changes often noted by owners can be categorized into four groups: mobility, activity level, grooming, and temperament. Mobility changes include reluctance to jump; not jumping as high; and changes in toileting behavior due to inability to climb into the litter box. Activity level changes manifest in decreased playing and hunting and a change in sleep patterns. Grooming changes may be noticed when the cat becomes more matted or unable to groom certain areas, and the claws may be overgrown because they cannot stretch out to "scratch/sharpen" claws. Changes in temperament are demonstrated by the cat hiding from owners or other pets in the house and seeming "grumpy" (Paster *et al*., 2005; Bennett and Morton, 2009). Many of these signs are again attributed to "old age" in the cat by the owner. Thus it is important for the technician to take a thorough history and ask open-ended questions that may help uncover otherwise overlooked signs of osteoarthritis in cats.

Diagnosing overweight/obesity is of the utmost importance and leads to diagnostic, curative, and preventive strategies that may be lost in the absence of a diagnosis. The first step to diagnosing overweight/obesity is consistent recording of both a body weight and BCS.

## Prevention

In dogs, risk factors for developing osteoarthritis include age, large or giant breeds, genetics, developmental orthopedic disease, trauma, and obesity. Risk factors for overweight/obesity in dogs include age, certain breeds, being neutered, consuming a semimoist, homemade, or canned food as their major diet source and consumption of "other" foods (meat or other food products, commercial treats, or table scraps). The radiographic prevalence of canine hip dysplasia, a leading cause of osteoarthritis in dogs, has been reported to be as high as 70% in golden retrievers and Rottweilers (Eby and Colditz, 2008). Golden Retrievers, Rottweilers, and Labrador Retrievers are overrepresented in the population of overweight/obese dogs.

Dogs found to be overweight at 9–12 months of age were 1.5 times more likely to become overweight adults (Kienzle *et al.*, 1998; Christensen *et al.*, 2007; Eby and Colditz, 2008; Anandacoomarasamy *et al.*, 2009). Owners of dogs at risk for obesity and osteoarthritis must be educated on the importance of lifelong weight management. The incidence and severity of osteoarthritis secondary to canine hip dysplasia can be significantly influenced by environmental factors such as nutrition and lifestyle (Impellizeri *et al.*, 2000; Kealy *et al.*, 2000; Burns, 2011). A long-term study has acknowledged that the prevalence and severity of osteoarthritis is greater in dogs with BCS above normal compared to dogs maintained at an ideal body condition throughout life. Over the lifespan of these same dogs, the mean age at which 50% of the dogs required treatment for pain attributable to osteoarthritis was significantly earlier (10.3 years, $P < 0.01$) in the overweight dogs as compared to the dogs with normal BCS (13.3 years).

Obesity is also a risk factor for the most common traumatic cause of osteoarthritis in dogs – ruptured cruciate ligaments. Overweight/obese dogs have a 2–3 times greater prevalence of ruptured cruciate ligaments compared to normal weight dogs. Understanding the correlation between maintaining their dog at a healthy weight and decreasing the risk of disease may be a powerful motivator for many owners.

In humans, the epidemic of obesity is largely attributed to changes in the availability, quantity, and composition of food and the decrease in the amount of physical activity needed for daily living. Physical activity levels of dogs often mirror their human companions. Owners should be encouraged to respond with play activities or praise rather than food rewards.

## Nutritional Management

Historically, the stress of excess weight on the skeletal system has been thought to be the primary offender in the pathophysiology and progression of osteoarthritis. Yet, adipose tissue is no longer considered simply a storage site for energy; rather it is now recognized as a multifunctional organ. Adipose tissue plays an active role in a variety of homeostatic and pathologic processes. Recent studies have documented that adipocytes secrete several hormones, including leptin and adiponectin, and produce a diverse range of proteins termed adipokines. Among the currently recognized adipokines is a growing list of mediators of inflammation: tumor necrosis factor α, interleukin-6, interleukin-8, and interleukin-10 (Towell and Burns, 2011). These adipokines are found in human and canine adipocytes. Production of these proteins is increased in obesity, suggesting that obesity is a state of chronic low-grade inflammation. Low-grade inflammation may add to the pathophysiology of a number of diseases commonly associated with obesity, including osteoarthritis. This also explains why somewhat small reductions in body weight can result in significant improvement in clinical signs.

## Nutrigenomics and Osteoarthritis

Nutritional supplementation of omega-3 fatty acids is a relatively new concept in the management of dogs with osteoarthritis. Recent studies provide high-quality data that show a diet with high levels of total omega-3 fatty acids and eicosapentaenoic acid (EPA) can improve the clinical signs of canine osteoarthritis. The use of a therapeutic canine food with higher levels of omega-3 fatty acids (specifically EPA) for the management of osteoarthritis has been supported by four randomized, double-blinded, controlled clinical trials using client-owned dogs (Fritsch *et al.*, 2010a, 2010b; Roush *et al.*, 2010a, 2010b). One 6-month study and two 3-month studies were conducted in US veterinary hospitals. In addition, a 3-month prospective study was carried out in two veterinary teaching hospitals. Overall, 500+ dogs with osteoarthritis were studied. Participating dogs were diagnosed with osteoarthritis based on history, clinical signs, and radiographic evidence. Dogs were fed either a typical commercial dog food or a test mobility food, which has higher concentrations of total omega-3 fatty acids and EPA and lower omega-6:omega-3 fatty acid ratios. At baseline and throughout the studies, subjective and objective veterinary evaluations were executed. Owners were also asked to subjectively evaluate their dogs throughout the studies.

These studies provide high-quality evidence that illustrate the benefits of incorporating a food with high levels of omega-3 fatty acids into the management of the pain of osteoarthritis in dogs. In normal canine cartilage there is a balance between synthesis and degradation of cartilage matrix. In arthritic joints damage to chondrocytes incites a vicious circle which culminates in the destruction of cartilage, inflammation, and pain. The mechanisms responsible for the demonstrated clinical benefits of omega-3 fatty acids include controlling inflammation and reducing the expression and activity of cartilage-degrading enzymes.

Cartilage degradation begins with loss of cartilage aggrecan and is followed by loss of cartilage collagens. This results in the loss of ability to resist compressive forces during movement of the joint. EPA is the only omega-3 fatty acid able to considerably decrease the loss of aggrecan in canine cartilage. It inhibits the upregulation of aggrecanases by blocking the signal at the level of messenger RNA (Caterson *et al.*, 2000; Caterson, 2004).

Inflammation is not only a vital reaction but it also plays an essential role in the pathophysiology of osteoarthritis. The polyunsaturated fatty acids (PUFAs) are critical components in the initiation and mediation of inflammation. Arachidonic acid (AA, 20:4$n$-6) and EPA (20:5$n$-3) act as precursors for the synthesis of eicosanoids, a significant group of immunoregulatory molecules that function as local hormones and mediators of inflammation. The amounts and types of eicosanoids synthesized are determined by the availability of the fatty acid precursor and by the activities of the enzyme systems that synthesize them. In most conditions, the principal precursor for these compounds is AA, although EPA competes with AA for the same enzyme systems. The eicosanoids produced from AA are proinflammatory and when produced in excess amounts may result in pathologic conditions. In contrast, eicosanoids derived from EPA promote minimal to no inflammatory activity (Wander *et al.*, 1997).

Eating foods which contain omega-3 fatty acids results in a decrease in membrane AA levels as omega-3 fatty acids replace AA in the substrate pool. This yields an accompanying decrease in the capacity to synthesize eicosanoids from AA. Studies have documented that inflammatory eicosanoids produced from AA are depressed when dogs consume foods with high levels of omega-3 fatty acids. In addition to their role in modulating the production of inflammatory eicosanoids, omega-3 fatty acids have a direct role in the resolution of inflammation. Resolution of inflammation is a progressive, active process involving a switch in the production of lipid-derived mediators over time. Proinflammatory products of omega-6 fatty acids metabolism (prostaglandin E2, prostaglandin E12, and leukotriene B4) are thought to initiate this sequence. AA-derived mediators foster the extravasation of inflammatory cells. With time and in the presence of sufficient levels of omega-3 fatty acids, a class shift occurs towards production of two families of pro-resolving omega-3-derived mediators – resolvins and protectins. These mediators serve as endogenous stop signals by preventing inflammatory cell recruitment, stopping "cell entry" and promoting resolution by removing inflammatory cells from the site. The identification of these two new families of omega-3-derived chemical mediators may clarify the mechanisms that underlie the many reported benefits of dietary omega-3 PUFAs. Absence of sufficient dietary levels of omega-3 fatty acids may contribute to "resolution failure" and perpetuation of chronic inflammation.

In cats with osteoarthritis two therapeutic foods are available for management of osteoarthritis in the United States. Hill's Prescription Diet® Feline j/d™ is available in the United States as well as Europe. The active ingredients include high levels of *n*-3 PUFAs (DHA), natural sources of glucosamine and chondroitin, methionine, and manganese. Just as in dogs, high levels of *n*-3 PUFAs control inflammation in cats. However, in cats DHA rather than EPA inhibits the aggrecanase enzymes responsible for cartilage degradation (Innes *et al.*, 2008; Burns, 2011). Natural sources of glucosamine and chondroitin increase proteoglycan production by chondrocytes and inhibit inflammatory mediators. Methionine and manganese enhance chondrocyte viability, provide building blocks, and act as a sulfur donor for the production of proteoglycans. Royal Canin Veterinary Diet® Mobility Support JS® Feline is available in the United States and Canada. The chief ingredient is green-lipped mussel, which contains anti-inflammatory constituents aimed at improving joint health. Other active ingredients are DHA, EPA, glycosaminoglycans, such as chondroitin sulfate, which are components of cartilage; an amino acid (glutamine), which is a precursor of glycosaminoglycans; and minerals important to maintaining healthy cartilage (i.e., zinc, copper, and manganese). The efficacy of therapeutic nutrition utilizing omega-3 fatty acids is supported by three studies (Frantz *et al.*, 2010; Fritsch *et al.*, 2010c; Sparkes *et al.*, 2010; Burns, 2011). The efficacy of therapeutic nutrition utilizing green-lipped mussel is supported by one published study (Lascelles *et al.*, 2010).

Research suggest therapeutic nutrition provides an effective and safe way to manage both dogs and cats with osteoarthritis. Foods with high levels of *n*-3 fatty acids have the dual value of controlling inflammation and pain while slowing progression of the disease by reducing cartilage degradation. Efficacy of therapeutic nutrition for osteoarthritis is supported by multiple clinical trials in arthritic pets.

### Developmental Orthopedic Disease

The goal of a feeding plan for pediatric pets is simple – to create a healthy adult. The specific objectives of a good feeding plan are to achieve healthy growth, optimize trainability and immune function, and minimize obesity and developmental orthopedic disease. Growth is a complex process involving interactions between genetics, nutrition, and other environmental influences. Nutrition plays a role in the health and development of growing pets and directly affects the immune system body composition, growth rate, and skeletal development.

Developmental orthopedic diseases (DODs) are a diverse group of musculoskeletal disorders that occur in growing puppies and may be related to nutrition. Canine hip dysplasia (CHD) and osteochondrosis are the most common musculoskeletal problems with a nutrition-related etiology. Specific nutritional factors that are thought to increase the risk of DOD in young dogs include:

- free choice feeding (excess energy consumption),
- feeding high energy foods (rapid growth), and
- excessive intake of calcium from food, treats and/or supplements (dietary imbalance) (Richardson *et al.*, 2010; Burns, 2014; Wortinger and Burns, 2015b).

Osteoarthritis secondary to DOD can be minimized by educating owners of young dogs to offer appropriate nutrition during the critical growth phase. All puppies whose adult weight is estimated to be ≥50 lb (23 kg) should be fed a growth food specifically formulated for large-breed dogs. As discussed earlier, maintaining an ideal BCS throughout life will decrease trauma to joints and the development of osteoarthritis.

## Patient Assessment

Pediatric pets should be assessed for risk factors before weaning to allow implementation of recommendations for appropriate nutrition.

A thorough history and physical evaluation are necessary. Special attention should be paid to large- and giant-breed puppies, breeds, and gender (including intact and neutered) at risk for obesity. Furthermore, growth rates and BCS provide valuable information about nutritional risks. Growth rates of young dogs are affected by the nutrient density of the food and the amount of food fed. It is important that puppies be fed to grow at an optimal rate for bone development and body condition rather than at a maximal rate. Growing animals reach a similar adult weight and size whether growth rate is rapid or slow. Feeding for maximum growth puts puppies at increased risk for skeletal deformities and has been found to decrease longevity in some species (Burns, 2014). In Labrador Retrievers, even moderate overfeeding resulted in overweight adults and decreased longevity.

The most practical indicator of whether or not a puppy's or kitten's growth rate is healthy is its BCS. Health care team members should be comfortable assessing BCS all patients; and with growing patients should reassess at least every 2 weeks to allow for adjustments in amounts fed and, thus, growth rates. Owners can and should be taught to assess body condition and are likely to become more aware of the appearance of a healthy growing puppy and kitten. Regularly assessing body condition provides immediate feedback about optimal nutrition.

## Key Nutritional Factors

The requirements for all nutrients are increased during growth compared with requirements for adult dogs. Most nutrients supplied in excess of that needed for growth cause little to no harm. However, excess energy and calcium are of special concern; these concerns include energy for puppies of small and medium breeds (for obesity prevention) and energy and calcium for puppies of large and giant breeds (for skeletal health). In addition, essential fatty acids can affect neural development and trainability of puppies.

### Energy

Energy requirements for growing puppies consist of energy needed for maintenance and growth. During the first weeks after weaning body weight is relatively small and the growth rate is high. Puppies use about 50% of their total energy intake for maintenance and 50% for growth. Gradually, the growth curves reach a plateau, as puppies become young adults. The proportion of energy needed for maintenance increases progressively, whereas the part for growth decreases. Energy needed for growth decreases to about 8–10% of the total energy requirement when puppies reach 80% or more of adult body weight. A puppy's DER should be about 3 times its RER until it reaches about 50% of its adult body weight (Wander *et al.*, 1997). Thereafter, energy intake should be about 2.5 times RER and can be reduced progressively to 2 times RER. When approximately 80% of adult size is reached, 1.8–2 times RER is usually sufficient.

These factors are general recommendations or starting points to estimate energy needs. Body condition scoring should be used to adjust these energy estimates to individual puppies.

Prevention of obesity is essential and should start at weaning. After puppies and kittens become overweight, it is challenging to return to, and maintain, normal weight. Too much food intake during growth may contribute to skeletal disorders in large- and giant-breed puppies. If the pet is overweight and/or obese and this is carried into adulthood, the risk for several important diseases is increased. These include hypertension, heart disease, diabetes mellitus, dyslipidemias, osteoarthritis, heat and exercise intolerance, and decreased immune function. Studies show that moderate energy and food restriction during the postweaning growth period reduces the prevalence of hip dysplasia in large-breed (Labrador Retriever) puppies and increases longevity in rats without hindering adult size (Richardson *et al.*, 2010; Burns, 2014; Wortinger and Burns, 2015b). Nonetheless, the pet may not receive enough energy and nutrients to support optimal growth if fed a food with a very low energy density and low digestibility. This may result in consumption of large quantities of the food, which can overload the gastrointestinal tract, resulting in vomiting and diarrhea. Health care team members should initiate monitoring of energy and food intake and body condition at an early age to help keep the pet at a healthy weight throughout life.

## Protein

Protein requirements of growing dogs differ from the requirements of adult dogs. During puppyhood, protein requirements are highest at weaning and decrease progressively until adulthood. Puppies 14 weeks and older, should receive at least the minimum recommended allowance for crude protein, which is 17.5% dry matter (DM). The recommended protein range in foods intended for growth in all puppies (small, medium, and large breed) is 22–32% DM. Most dry commercial foods marketed for puppy growth provide protein levels within this range (Burns, 2014; Wortinger and Burns, 2015c).

Protein levels above the upper end of this range have not been shown to be detrimental but are well above the level in bitch's milk. Protein requirements of growing dogs differ from those of adults. An important difference is that arginine is an essential amino acid for puppies, whereas it is only conditionally essential for adult dogs. Foods formulated for adult dogs should not be fed to puppies (Burns, 2014). Although protein levels may be adequate, energy levels and other nutrients may not be balanced for growth.

## Fat

Growing dogs have an estimated daily requirement for essential fatty acids (linoleic acid) of about 250 mg/kg body weight, which can be provided by a food containing between 5% and 10% DM fat. Research has shown that docosahexaenoic acid (DHA) is essential for normal neural, retinal, and auditory development in puppies. Inclusion of fish oil as a source of DHA in puppy foods improves trainability and should be considered essential for growth. The minimum recommended allowance for DHA plus EPA is 0.05% DM; EPA should not exceed 60% of the total. Thus, DHA needs to be at least 40% of the total DHA plus EPA, or 0.02% DM (Richardson *et al.*, 2010).

When feeding young pets, we must remember that fat contributes greatly to the energy density of a food and excessive energy intake can cause overweight/obesity and DOD. The minimum recommended allowance of dietary fat for growth (8.5% DM) is much less than that needed for nursing, but more than is needed for adult maintenance (5.5% DM). In order to deliver a DM energy density between 3.5 and 4.5 kcal/g, 10–25% DM fat is necessary. This range of dietary fat is recommended from postweaning to adulthood (Richardson *et al.*, 2010).

The principal functions of dietary fat are to act as:

- a source of essential fatty acids,
- a carrier for fat-soluble vitamins, and
- a concentrated source of energy.

## Calcium and Phosphorus

Although growing dogs need more calcium and phosphorus than adult dogs, the health care team must remember to educate owners that the minimum requirements are relatively low. Puppies have been successfully raised when fed foods containing 0.37–0.6% DM calcium and 0.33% DM phosphorus (Richardson *et al.*, 2010).

Large- and giant-breed puppy foods should contain 0.7–1.2% DM calcium and 0.6–1.1% phosphorus. Foods with a calcium content of 1.1% DM provide more calcium to puppies just after weaning than if bitch's milk is fed exclusively. Small- to medium-sized breeds are less sensitive to slightly overfeeding or underfeeding calcium; thus the level of calcium in foods for these puppies can range from 0.7 to 1.7% DM, (0.6–1.3% phosphorus) without risk. The phosphorus intake is less critical than the calcium intake, provided the minimum requirements of 0.35% DM are met and the calcium:phosphorus ratio is between 1:1 and 1.8:1. For large- and giant-breed dogs, the calcium:phosphorus ratio should be between 1:1 and 1.5:1 (Richardson *et al.*, 2010).

### Digestibility

Puppies that are fed foods with decreased energy density and decreased digestibility will need to eat larger amounts of food to achieve growth. Consequently, this increases the risk of flatulence, vomiting, diarrhea, and the potential development of a "pot-bellied" appearance. As a result, foods recommended for puppies should be more digestible than typical adult foods. An indirect indicator of digestibility is energy density. Foods with a higher energy density are likely to be more digestible (Richardson *et al.*, 2010; Burns, 2014).

### Carbohydrates

Although no specific level of digestible (soluble) carbohydrates exists for growing puppies, it is recommended that the level of digestible (soluble) carbohydrates around 20% DM may optimize health.

## Conclusion

Successful treatment and prevention of musculoskeletal disease conditions require a comprehensive approach which includes preventive measures and a multimodal management program. Clinical signs of musculoskeletal diseases are often not obvious on examination, especially early in the disease process. Although signs of overweight/obesity are readily apparent they are often overlooked or dismissed as inconsequential. Documenting a diagnosis of overweight/obesity is critical to the management of these disease conditions. Diagnosing overweight/obesity requires consistent recording of both a body weight and BCS. Early diagnosis of osteoarthritis and DOD enables early intervention, which in turn often improves the long-term outcome for the patient. Consistent use of a thorough, disease-specific history may raise awareness of subtle changes early in the course of osteoarthritis and DOD. Successful management of osteoarthritis/DOD and obesity requires nutritional intervention.

Rehabilitation programs are designed to be part of a multimodal approach. One important part of the rehabilitation of veterinary patients is nutrition. Health care teams should have knowledge of nutrition and specific nutrients which play a role in certain disease conditions. Rehabilitation incorporates a number of treatment modalities. A nutritional plan should be part of each patient's rehabilitation therapy and should continue following successful completion of the rehabilitation program. Reassessment is required until nutritional and rehabilitative goals are met.

## References

Ahima RS (2006) Adipose tissue as an endocrine organ. *Obesity* 14(Suppl 5):242S–249S.

Anandacoomarasamy A, Fransen M, and March L (2009) Obesity and the musculoskeletal system. *Curr Opin Rheumatol* 21:71–77.

Anandacoomarasamy A, Caterson I, Sambrook P, *et al.* (2008) The impact of obesity on the musculoskeletal system. *Int J Obes (Lond)* 32:211–222.

Bennett D and Morton C (2009) A study of owner observed behavioural and lifestyle changes in cats with musculoskeletal disease before and after analgesic therapy. *J Feline Med Surg* 11(12):997–1004.

Burkholder WJ and Toll PW (2000) Obesity. In *Small Animal Clinical Nutrition*, 4th edn. Walsworth Publishing Co., Missouri, MO, pp. 402–426.

Burns KM (2011) Are your patients suffering in silence? Managing osteoarthritis in pets. *NAVTA J* Convention Issue:16–22.

Burns KM (2014) Pediatric nutrition: Optimal care for life! *NAVC Proceedings*, Orlando, FL, January 18–22.

Caterson B, Flannery CR, Hughes CE, *et al.* (2000) Mechanisms involved in cartilage proteoglycan catabolism. *Matrix Biol* 19:333–344.

Caterson G (2004) Omega-3 fatty acids-incorporation in canine chondrocyte membranes. Unpublished data. Cardiff University, Wales, UK.

Christensen R, Bartels EM, Astrup A, and Bliddal H (2007) Effect of weight reduction in obese patients diagnosed with knee osteoarthritis: a systematic review and meta-analysis. *Ann Rheum Dis* 66:433–439.

Eby J and Colditz G (2008) Obesity/overweight: prevention and weight management. In *International Encyclopedia of Public Health* (eds. S Quah and K Heggenhougen). Elsevier, St. Louis, MO, pp. 602–609.

Frantz NZ, Hahn K, MacLeay J, *et al.* (2010) Effect of a test food on whole blood gene expression in cats with appendicular degenerative joint disease. *J Vet Intern Med* 24:771.

Fritsch D, Allen TA, Dodd CE, *et al.* (2010a) Dose-titration effects of fish oil in osteoarthritic dogs. *J Vet Intern Med* 24:1020–1026.

Fritsch DA, Allen TA, Dodd CE *et al.* (2010b) A multi-center study of the effect of dietary supplementation with fish oil omega-3 fatty acids on carprofen dosage in dogs with osteoarthritis. *J Am Vet Med Assoc* 236:535–539.

Fritsch D, Allen TA, Sparkes A, *et al.* (2010c) Improvement of clinical signs of osteoarthritis in cats by dietary intervention. *J Vet Intern Med* 24:771–772.

German, AJ, Holden, SL, Moxham, GL, *et al.* (2006) A simple, reliable tool for owners to assess the body condition of their dog or cat. *J Nutr* 136:2031S–2033S.

Impellizeri JA, Tetrick MA, and Muir P (2000) Effect of weight reduction on clinical signs of lameness in dogs with hip osteoarthritis. *J Am Vet Med Assoc* 216:1089–1091.

Innes J, Gabriel N, Vaughan-Thomas A, *et al.* (2008) *Proceedings Hill's Global Mobility Symposium*, 22–26.

Kealy RD, Lawler DF, Ballam JM, *et al.* (2000) Evaluation of the effect of limited food consumption on radiographic evidence of osteoarthritis in dogs. *J Am Vet Med Assoc* 217:1678–1680.

Kienzle E, Bergler R, and Mandernach A (1998) A comparison of the feeding behavior and the human-animal relationship in owners of normal and obese dogs. *J Nutr* 128:2779S–2782S.

Laflamme DP (2006) Understanding and managing obesity in dogs and cats. *Vet Clin Small Anim Pract* 36:1283–1295.

Lascelles BDX, DePuy V, Thomson A, *et al.* (2010) Evaluation of a therapeutic diet for feline degenerative joint disease. *J Vet Intern Med* 24:487–495.

Lascelles BD and Robertson SA (2010) DJD-associated pain in cats: what can we do to promote patient comfort? *J Feline Med Surg* 12:200–212.

Paster ER, LaFond E, Biery DN, *et al.* (2005) Estimates of prevalence of hip dysplasia in Golden Retrievers and Rottweilers and the influence of bias on published prevalence figures. *J Am Vet Med Assoc* 226(3):387–392.

Richardson DC, Zentek J, Hazewinkel HAW, *et al.* (2010) Developmental orthopedic disease in dogs. In *Small Animal Clinical Nutrition*, 5th edn. (eds. M Hand, CD Thatcher, R Remillard, *et al.*). Mark Morris Institute, Topeka, KS.

Rosenthal M (2007) Obesity in America: Why Brune and Bessie are so heavy and what you can do about it. *Vet Forum* 24:26–34.

Roush JK, Dodd CE, Fritsch DA, *et al.* (2010a) Multicenter practice assessment of the effects of omega-3 fatty acids on osteoarthritis in dogs. *J Am Vet Med Assoc* 236(1):59–66.

Roush JK, Cross AR, Renberg WC, *et al*. (2010b) Evaluation of the effects of dietary supplementation with fish oil omega-3 fatty acids on weight bearing in dogs with osteoarthritis. *J Am Vet Med Assoc* 236(1):67–73.

Sparkes A, Debraekeleer J, and Hahn KA (2010) An open-label, prospective study evaluating the response to feeding a veterinary therapeutic diet in cats with degenerative joint disease. *J Vet Intern Med* 24:771.

Thatcher CD, Hand MS, and Remillard R (2010) An iterative process. In *Small Animal Clinical Nutrition*, 5th edn. Mark Morris Institute, Topeka, KS, pp. 3–21.

Towell TL and Burns KM (2011) Multimodal management of osteoarthritis. *NAVC Proceedings*, Orlando, FL, NAVC Conference 2011, January 15–19.

Wander RC, Hall JA, Gradin JL, *et al*. (1997) The ratio of dietary (n-6) to (n-3) fatty acids influences immune system function, eicosanoid metabolism, lipid peroxidation and vitamin E status in aged dogs. *J Nutr* 127:1198–1205.

Wortinger A and Burns K (2015a) *Nutritional Management of Disease for Veterinary Technicians and Nurses*. Wiley Blackwell, Ames, IA.

Wortinger A and Burns KM (2015b) Weight management. In *Nutrition for Veterinary Technicians and Nurses*. Blackwell, Ames, IA, pp. 159–167.

Wortinger A and Burns KM (2015c) Pediatric nutrition. In *Nutrition for Veterinary Technicians and Nurses*. Blackwell, Ames, IA, pp. 159–167.

Wortinger A and Burns KM (2015d) Musculoskeletal. In *Nutrition for Veterinary Technicians and Nurses*. Blackwell, Ames, IA, pp. 159–167.

# 9

# Motivating Your Patient

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CHAPTER MENU



## Introduction: Why we Need to Motivate the Rehabilitation Patient

Patients that need rehabilitation are often painful. It is also likely that the patient has undergone multiple previous examinations for the issue that the patient is presenting for, before the rehabilitation examination and treatment even occurs. This means that the painful area has been palpated repeatedly. The natural guarding of an injured body part, which is part of protecting an injury while it heals, can be compounded and amplified by the repeated experiences of physical examination. We expect a lot of our patients, often without giving them coping mechanisms. Most veterinary patients have not been fully desensitized to a routine, brief veterinary physical examination, even after multiple visits to the primary care veterinarian. The rehabilitation examination is a much more extensive than a routine examination and hence is a prolonged process with a great deal of patient handling.

A relaxed patient is more compliant and easier to examine and diagnose; muscle tension is reduced and pain signs are easier to discern

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when the overall stress level of a patient is low. It is mutually beneficial for the patient and for the veterinary team to have the examination and subsequent therapies in a low-stress, high-reward environment for the patient.

After examination and diagnosis, a rehabilitation prescription is made. This prescription can involve both in-clinic therapies and daily in-home therapies. A relaxed environment for the initial evaluation process, and, consequently, a relatively calm patient, also helps the client to fully participate in learning about the pet's condition and the at-home and in-clinic therapy plan without distractions of stress and concern for the pet. Both active and passive exercises may be prescribed for the patient. A motivated patient will comply with the plan, whereas forcing an exercise by using restraint without motivation often leads to increasing reluctance from the patient. Reluctance reduces both patient and client compliance. The rehabilitation team should aim for as much happy patient engagement as possible to achieve the best results. Building and maintaining a good relationship with each patient is important for the rehabilitation team in clinic, but vital for success with at-home exercises.

## Understanding Animal Psychology

It is important to appreciate that reinforcement and punishment are defined by their effect, not by the intent of anyone who may be delivering them. When discussing positive and negative aspects of a handling or motivation technique, it should only be about how that approach is received by the patient. It is common to assume that since no harm is meant, the procedures are not perpetuating fear in the patient. Teaching this approach to the client starts with understanding and explaining the situation from the animal's point of view. There are many resources that make understanding canine body language easier. For example, in Doggie Drawings (www.doggiedrawings.net), drawings starring a Boston Terrier named Boogie illustrate important information such as "How Not to Greet a Dog" and "Doggie Language" in an informal and entertaining way. The American Association of Feline Practitioners has created an online resource for cat behavior that covers many topics (www.catvets.com). "Pleasant veterinary visits for cats" points out that "clients are more willing to obtain regular veterinary care, including more extensive preventive and therapeutic care, if the visits are pleasant. Calm, relaxed cats enable us to perform more thorough physical examinations and enable clients to better focus on our recommendations" (American Association of Feline Practitioners, 2004).

Helping clients to understand how to better read their pet's body language leads to being able to apply that knowledge, and then change their approach to their pet during an exercise as needed in order to improve comfort and compliance. This will foster a successful relationship long term. It is important for us, as professionals, to use correct terminology when discussing and teaching behavior modification and training. For a glossary of terms, see Box 9.2.

## Classical Conditioning Versus Operant Conditioning

### Reinforcing Behaviors

The most well-known example of classical conditioning is Ivan Pavlov's experiments with salivating dogs (Pavlov, 1928). Present food, and the dog salivates. Ring a bell, and the dog does not salivate. Ring a bell then present food, and the dog salivates. Ring the bell, and the dog now salivates. Pavlov effectively used classical conditioning to associate a ringing bell with food (Overall, 2013). We often use this to our advantage when working with our patients in the clinic. Many of the patients that we treat are fearful of new surroundings, strangers, and new activities. When we pair a primary reinforcer – something innately reinforcing to the dog (typically food) – with a neutral (uninteresting)


Figure 9.1 Frozen treats take time to consume, allowing the animal to work at the reward.

or stressful experience, we can create a positive association and change how the dog perceives the situation for the present, and how he or she will perceive the same situation in the future. It is important to note that we should make every effort to create positive experiences, not just neutral experiences. At our clinic, Twin Cities Animal Rehabilitation and Sports Medicine, a few treats utilized are freeze-dried meat (which is also popular with our feline patients), peanut butter, and frozen peanut butter yogurt cups (Figure 9.1). For the increasing number of dogs with food intolerances, hydrolyzed (hypoallergenic) treats are available as well.

**Real Life Example**
A client was unwilling to use food rewards in clinic.

Client: "Sparky doesn't need treats to behave."

Technician: "We want to build the best association with us, the building, and the equipment we can. Using food helps us to reinforce calm, cooperative behavior."

Client: "He needs to listen because I said so."

Technician: "We use food to see how willing Sparky is to do exercises for us. If we know Sparky likes food, and knows how to earn food but will not sit, I will ask for other behaviors (down, stand) first. Then ask for sit. If he will make the other movements, but still shows hesitation to sit, this can help us to identify pain and understand where it hurts so that we can help Sparky to recover."

## Emotional Response – Creating Associations with the Environment

Where Pavlov is known for his work with the bell, B.F. Skinner is known for the four quadrants of operant conditioning (Skinner, 1938). Skinner's principles were that desired behaviors should be reinforced; behaviors can be built in incremental steps; and that immediate reinforcement generally provides a better learning experience than delayed reinforcement. These principles have been used extensively in training. Operant conditioning breaks learning down into four areas or subsets. In each area we evaluate our training/motivating based on whether we have successfully increased or decreased the frequency of a behavior (Martin and Martin, 2009) (Box 9.1).

Positive reinforcement is often used, and usually without thinking too much about it.

**Box 9.1 Reinforcements**

| | Positive (add) | Negative (remove) |
|---|---|---|
| Reinforcement (desirable) | Add something desirable to increase the frequency of a behavior | Remove something aversive to increase the frequency of a behavior |
| Punishment (undesirable) | Add something undesirable to decrease the frequency of a behavior | Remove something desirable to decrease the frequency of a behavior |

*Source:* Adapted from Martin and Martin (2009).

Successful influence on behavior using this technique generally occurs, but we can always try to do a better job of communicating with our patient more clearly. A key piece, often missing when motivating a patient, is looking closely at what exact behavior or action is being reinforced. This requires precise attention to detail. Dogs are uniquely attuned to small motions; for example, the handler may be rewarding a "behavior chain" of brief contact of one foot on a balance disc with a step back off again, when the goal is to reinforce standing still with both front feet on a balance disc.

Improving the timing of rewards, typically by adding in a "secondary reinforcer," will shorten the amount of training time. The most common secondary reinforcer is a "marker" (a unique way to mark a behavior), such as a clicker, verbal tongue clicks, verbal yes, or other less commonly heard word (avoid "okay"). The purpose of the secondary reinforcer is to give the handler more time between desired behavior and getting around to giving the primary reinforcer (reward). A secondary reinforcer helps to give the animal clarity about the exact behavior (e.g., limb placement) that is desired and so being reinforced. The animal knows that a reward will follow the secondary reinforcer. The patients typically learn about a secondary reinforcer without too much dedicated training if there is consistency with the type of marker used. Mark the behavior, then deliver the reward. Practice mark, followed by reward, for multiple new behaviors to build value for the marker. Since both hands are usually needed for restraint and carrying out exercises, using a secondary reinforcer that is a verbal marker will help an animal to identify the behavior being rewarded, and this will give the handler time to access, then deliver, the treat reward.

In cases where a verbal marker is utilized, remember that inflection (the tone and pitch of your voice as it says the word) can impact the effect. For this reason, a tongue click may be preferred over a verbal "yes." One of the rules of good training is to always pair the secondary reinforcer with food or it will quickly lose value. Using these techniques while the client is in clinic with the patient allows time for observation and learning techniques to apply at home, and time for answering questions if needed (Box 9.2).

**Box 9.2 Key terms used for reinforcement**

- *Reinforcer* – Any consequence that causes the preceding behavior to increase (e.g., treats)
- *Behavior chain* – Specific sequence of discrete responses, each associated with a stimulus
- *Primary reinforcer* – Primary reinforcers are biological (e.g., food, pleasure from a toy)
- *Secondary reinforcer* – A conditioned reinforcer; the dog knows something good is coming
- *Marker* – A unique way to mark behavior (e.g., "yes!")
- *Luring* – Using a reward to increase attention and interest (e.g., to initiate motion)
- *Conditioning* – Associations between stimuli and response

Working to make positive or neutral associations with the environment in the clinic can include more than just the use of reinforcers. Maintaining a quiet environment is extremely important for some patients. Patients who are reactive to noise can react to something as small as a beep from an ultrasound machine. It is possible to get the sounds turned off on some ultrasound and e-stim machines. Shockwave units can make a lot of sound. The use of ear mufflers can help in patients that are not sedated for therapy, and can also help lightly sedated patients (Figure 9.2). Sometimes a white noise machine can help reduce sounds outside the treatment room. The white noise of some underwater treadmill units can dull general clinic noise in some cases. All staff members should be cognizant of raising their voices, even in laughter. Calming music can help some patients to relax, and there are specific tempos of music made to calm dogs (e.g., Through A Dog's Ear™) available in CD

Figure 9.2 Patient undergoing shockwave therapy. He is wearing ear muffs and licking a frozen treat.

Figure 9.3 A low matted table for examination and therapy, the table is 40 cm high.

or MP3 format. It is recommended to have individual stereos/speakers in each room to allow for volume control rather than having music playing at the same level throughout the clinic. The importance of comfortable, traction flooring should not be underestimated; a patient will feel more relaxed if they are better able to grip the floor. Examination tables, if used, should have traction; low matted tables are recommended (Figure 9.3).

## Understanding Body Language

Evaluating stress while in the clinic can be challenging. A stress level scale developed by Dr. Karen Overall is a helpful tool to accurately assess the current stress level for a dog while in clinic (Overall, 2013). Knowing how to properly evaluate the behavior of a patient is going to help to foster the best relationship, so that we can meet our goals. We can also provide some assignments for the client to work on these handling situations in a positive way at home.

The most basic signs of stress are yawning, lip licking, or blinking. The most severe is biting for carnivores (biting and kicking for equids and ungulates). Most patients fall somewhere in the middle of this spectrum; they are not completely comfortable with what is going to happen, but are not predisposed to bite. Dr. Overall suggests taking note of body posture, tail posture, ear posture, gaze, pupils, respirations, lips, activity, and vocalization. If we take the time to learn about what the patient behaviors mean, we can make decisions about how thorough and lengthy the hands-on portion of each visit is (Figures 9.4 and 9.5). We can quickly discern when a patient needs a break so that stress (cortisol) levels can drop back to a more neutral state. In cases of rising stress with danger signs, we can evaluate whether we need to postpone the examination, or restrain the patient for safety of personnel; the best option for all concerned being to postpone the examination/therapy when patient stress is reaching concerning levels.

## Building Motivation and Value Through Food Rewards

As discussed previously, food is a powerful primary reinforcer for our patients. Utilizing this effectively includes gathering information about diet and feeding rituals in the home. Two of the most common reasons for lack of food motivation are free feeding at home and weight control issues. Free feeding is when food is left out always for the pet. This can be a measured amount or filled at will. The measured amount still often means

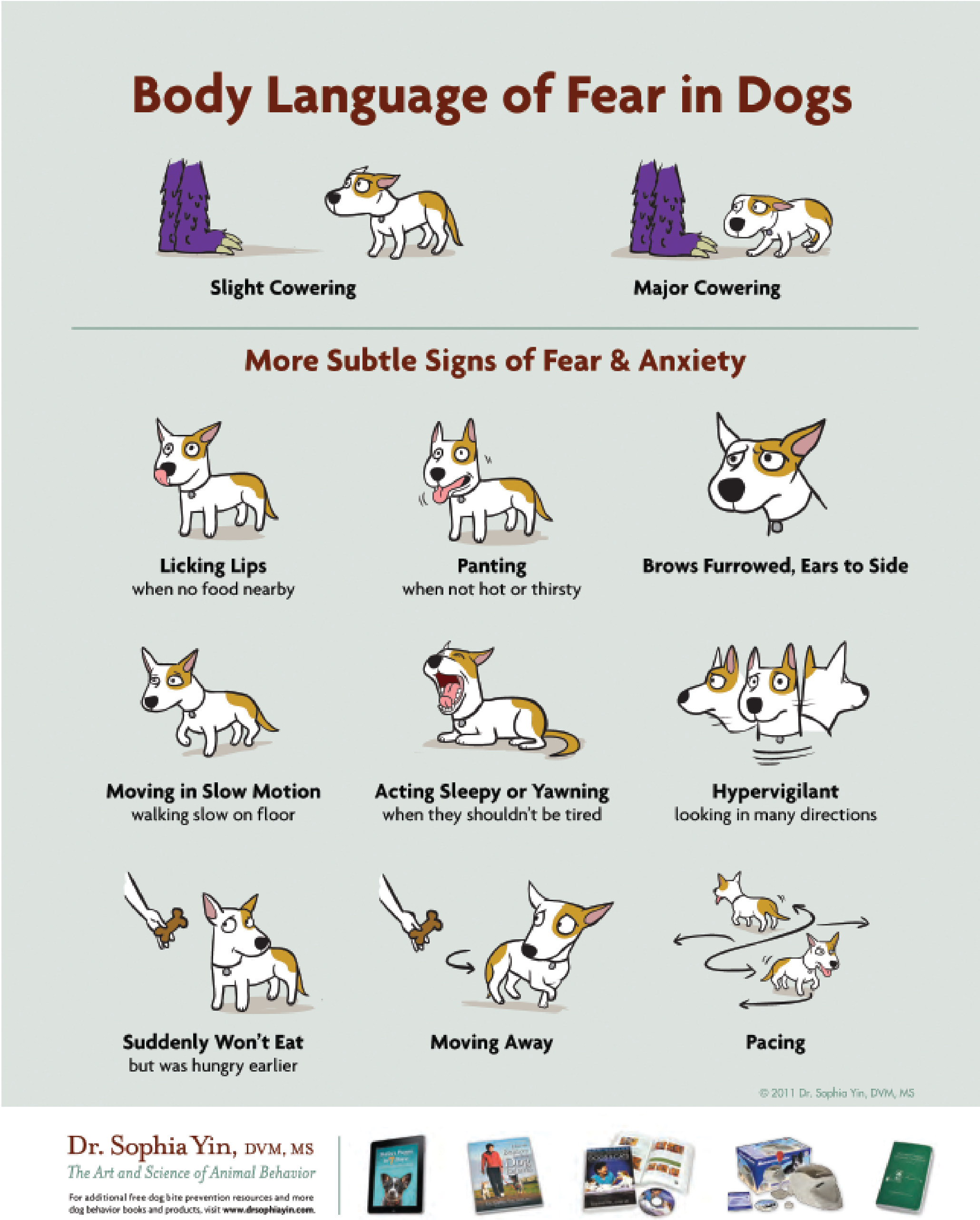


Figure 9.4 Poster showing dog behavioral postures. *Source:* Courtesy of Dr. Sophia Yin.

no portion control. Free feeding makes our primary reinforcer (food) far less rewarding for the dog, as it is always available; there is no drive to eat from a human hand. Many free-fed dogs are also overweight, so by addressing one problem, we address both. Food is such a powerfully useful tool and it is often overlooked. Help the client to set up a feeding schedule that works for everyone. For dogs with a necessity for low caloric

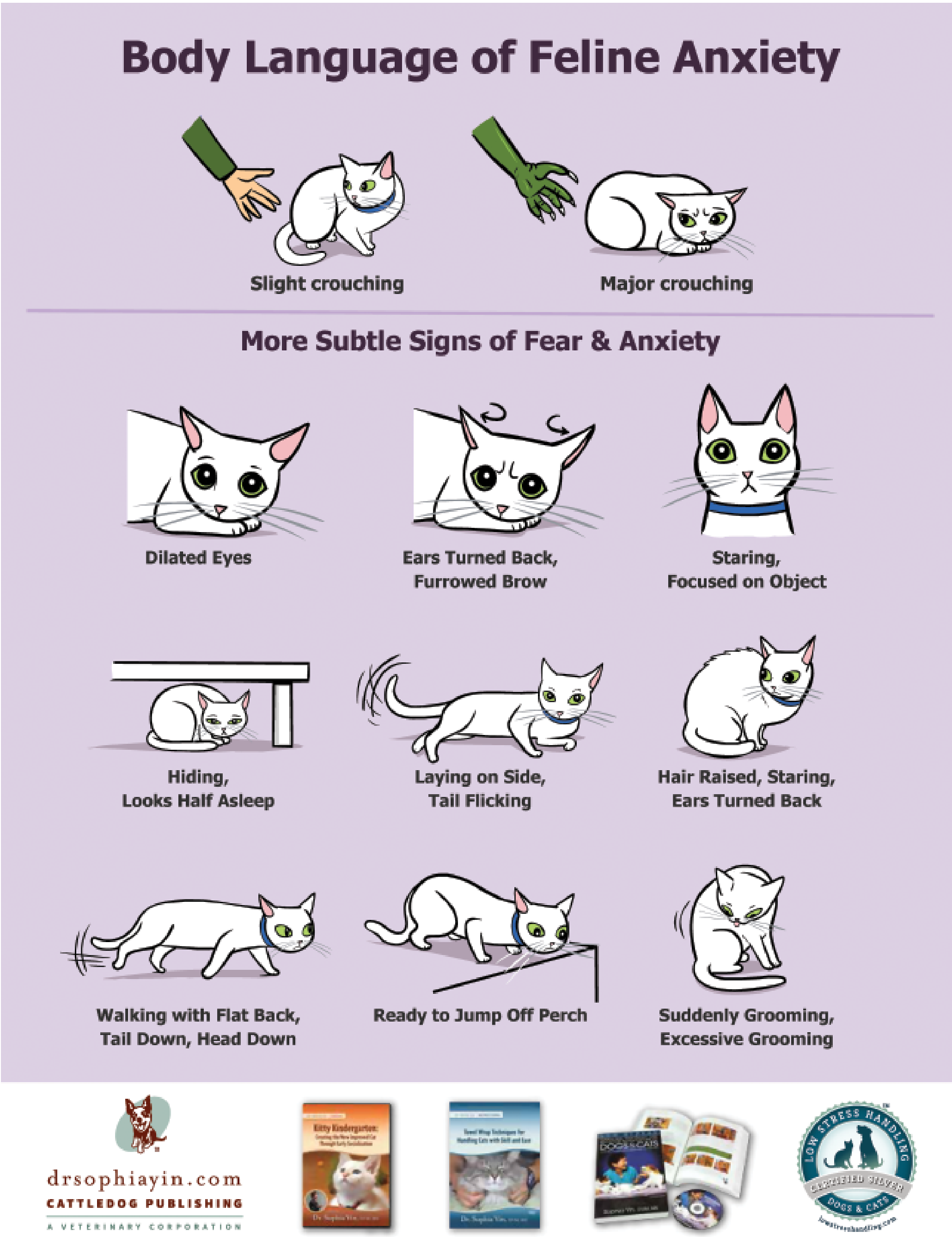


Figure 9.5 Poster showing cat fear and anxiety postures. *Source:* Courtesy of Dr. Sophia Yin.

intake, a common recommendation is to use meals as treats. By teaching the dog to work for meals, a lot is accomplished; when appetite is high, so is drive and attention to the reward. Using the normal diet as reinforcement while in familiar environments allows for a wider variety of high-value reinforcement in more stimulating (usually also more stressful) environments. Reserving a variety of treats specifically for therapy appointments can help the whole process to proceed more smoothly. For cats, very high-value foods with strong odors (freeze-dried meat, baby food) can help to stimulate a cat to eat; for example, move to eat while stretching front legs up onto a raised surface, so elongating the spine and extending the hips (Figure 9.6). Many horse clients and trainers use treats. For both cats and horses, it is even more important than it is for dogs to have a treat that takes multiple bites or licks to consume as these patients tend to gain less reinforcement from interaction with people.

A positive relationship is fostered if the goal of motivation and training is cooperation. Prescribed rehabilitation exercises provide an opportunity for training, thereby providing the patient with mental stimulation and the client with strengthening of the human–animal bond. Patients who are undergoing rehabilitation are usually exercise restricted. This can lead to unwanted behaviors due to boredom and excess energy. Training requires concentration from the animal (as well as the handler), this mental activity and stimulation causes some fatigue, and so decreases boredom.

Figure 9.6 A feline patient being lured into stretch using freeze-dried meat. Note his left rear extends less. The patient has had a femoral head excision.

Replacing the normal food bowl with a work-to-eat toy (also known as a food toy) is another great way to encourage mental stimulation and avoid boredom. For horses, work-to-eat toys such as the Amazing Graze™ treat toy (www.HorsemensPride.com) can help to alleviate the boredom of stall confinement. The rehabilitation team needs to be cognizant of the type of toy and the effort and motions required to release food from the toy. Choosing an appropriate work-to-eat toy needs to be taken into consideration. A toy that requires a lot of movement with the patient concentrating on rolling the toy, instead of on their gait compensations, may result in relative overload of an injury. The disabilities of the patient need to be considered, and the team should be sure to counsel the client about providing the work-to-eat toy in a confined area.

## Luring, Shaping, and Marking Behaviors

Luring is the motivating technique that is used most often, as most patients can be motivated by food. Having an assortment of high-value reinforcers (treats and fun toys), as well as instructing clients to bring known favorite treats from home can help. Even the most outgoing and typically social dogs can be unsettled by new experiences. The rehabilitation team should be creative with the treats, but as mentioned previously, reserve the exciting stuff for situations where it will be harder to motivate due to distraction and stress (for example a treatment such as dry needling of myofascial trigger points, which causes some discomfort). A helpful recommendation is to make up a bag of goodies to be kept in the refrigerator or freezer that combines a few really tasty snacks and some regular food. An example snack "grab bag" ready to go with everything cut up into small pieces would contain: cubes of cheese, strips of boiled chicken, regular dry food, and a commercially available soft treat.

Although it may seem to be a skill that most dogs are born with, teaching an animal to "lure" is invaluable. Luring is when the animal follows the treat (Martin, 2009b) to perform a desired behavior. Most clients are familiar with luring without realizing, but may not be maximally successful. It is important to feed often during the luring process, to keep the patient focused more on the food than on the goal behavior (a desired motion, or simply remaining still during a therapy). For this reason, treats that can be nibbled on or something that can be licked at, providing a slow, continuous reward, are great options. When teaching a dog to lure, start with small steps, do not have the complete behavior as the goal; instead, reward small behaviors. Many dogs tend to sink down to the ground instead of standing still when manipulated. When we need a dog to stand for examination, luring into that position would be a multistep process. First let the dog nibble from their current, undesired, position. Next bring the treat up and forward just a little so the dog continues to be engaged. Finally reward the full stand with a rapid succession of treats in a short "burst" (Figure 9.7). The posture is maintained by continued intermittent rewards.

Teaching a horse or a cat to lure is a very similar process, but motivation may differ between individuals.

Luring requires focus on the part of the animal. Overly exuberant dogs are approached in a similar manner to shy dogs so that stress/unfocused energy does not distract from the food-luring process. Present the food to the patient without eye contact, and with your body turned to the side, if the dog acknowledges you but does not take from your hand, then gently toss a treat across the floor. Aim to toss the treat behind the dog, not in their face. Once the patient recognizes you as the source of the treat, they are more likely to accept a treat given from your hand. These non-assertive behaviors by rehabilitation team members will keep the dog focused on the treats and avoid stressors, overexcitement, and other distractions.

Figure 9.7 Luring a patient from a sit into a standing position for examination

## Simple Games: Set Up for Success

As mentioned previously, a patient that can be lured into different positions is much easier to work with. Starting with behaviors that are easily lured helps keep patient and owner motivated with success. Having the patient follow a food luring head up, head down, body in a circle, into a sit, or into a stand are some basic behaviors that most dogs do on a regular basis. Cats will stretch and move their body in response to luring too. Breaking the final behavior down into smaller, incremental steps helps to set up both client and patient for success. If the end goal is to have the patient lie in lateral or in sternal recumbency, it is necessary to think of the behaviors that build up to that. Clients who get involved in the whole process of making the prescribed rehabilitation therapies into a game tend to be more successful in maintaining, or in many cases, strengthening their human–animal bond and in completing the rehabilitation goals.

## Having Intent and Understanding Trust

Leading by example is a powerful way to teach clients about building and maintaining trust with their animal, and their animal's trust of the rehabilitation team. During an initial consultation, while the veterinarian is learning about the patient's history, the technician can be building a relationship with the patient. The luring process can initially be time consuming (e.g., tossing treats to a shy or fearful patient and slowly building trust), so beginning the process during the time taken to record patient history and during the time taken to make initial observations about the patient before handling (postures, gait, and transitions, etc.) is efficient. This approach means that by the time the patient needs to be handled, they have had some exposure to the luring/reward process.

## Home Exercises – Knowing When to Stop

Many clients who bring a pet in for rehabilitation are unaware of subtle signs of pain. Vocalizing is typically the only sign recognized by clients, so education about pain-related behaviors and understanding pain level at home is an important part of the initial evaluation process. Every exercise prescribed by the veterinarian must be performed with comfort in mind. Improper application of technique, changes in patient status/function, or fatigue can be the difference between an exercise going smoothly without pain and one being met with resistance due to pain and stress. Pain and fatigue levels can vary day to day depending on other activity level and additional compounding factors. Counseling clients in typical pain and stress-related body language will help to set the stage for knowing when to stop (Figure 9.8). A patient who is averting eyes, shifting away, lip licking, or pulling ears back is showing signs of pain and discomfort; these signs occur at relatively mild levels of discomfort, and long before vocalizing will happen. Proceeding with more caution and modifying the exercise to decrease physical pressure (e.g., limiting end-range joint motions) will not only keep the patient more comfortable physically, but it will help to preserve the relationship between patient and client. If in doubt, the client should stop the exercise and it should be re-evaluated with the client in the rehabilitation clinic. Video footage of the exercise and associated behaviors at home may also be helpful. Hesitation is often the first sign of fatigue, but can also be the first sign of discomfort. Be sure to counsel about rest and about not pushing through an exercise if the patient's willingness decreases.

Rehabilitation therapy can cause mild transient discomfort but should not be an uncomfortable process beyond this. In the veterinary world, we rely on patient compliance and a breakdown of this compliance can mean an


Figure 9.8 International Veterinary Academy of Pain Management (IVAPM) pet pain awareness poster for client education, 2016. *Source:* Courtesy of IVAPM.

end to therapy; we cannot force our way through a patient's pain by reasoning with the patient about long-term gains being worth the discomfort. We aim to avoid philosophies espoused in human therapy and sports training such as "no pain, no gain." Instead we focus on teaching cooperation and compliance to develop a willing partnership between patient, client, and the rehabilitation team.

## Problem Solving and Dealing with Challenging Patients

### Case Examples

#### Case No. 1: Human Reactive and Handler Aggressive

A 7-year-old German Shepherd presented for work-up of a forelimb lameness. She had been rescued from a reportedly abusive home and exhibited fear-based anxiety that led to aggression if not kept under threshold. She was not acclimated to a muzzle and had low level of treat motivation. For her initial evaluation, she had received anti-anxiety medications, essential oils, and herbal calming supplements prior to arriving at the rehabilitation clinic. The client was initially insistent on handling the dog. The patient would take treats when tossed near to her and progressed to taking treats out of the technician's and veterinarian's hands; however, when attempts were made to handle her forelimbs she became very anxious, demonstrated by vocalizing and showing her teeth.

A soft cone was placed around her head by the client, followed by feeding treats rapidly. She was then restrained as lightly as possible while the client fed her, but she would not always eat. Forelimb examination was as brief as possible. It was determined that diagnostic ultrasound was needed to obtain a full picture of the cause of the identified shoulder pain. A low-stress plan was made for the next visit. It was determined that removing the client from the room decreased the stress of the dog. In addition to the medications, oils, and supplements she received prior to her first visit, we played calming music (Through A Dog's Ear™) and worked as efficiently as possible.

It was noted that she settled better with a hands-off approach and would take tossed food, but she was unfortunately not trustworthy being unrestrained. Throughout the diagnosis and treatment of her front limb injury, we discovered that she was much more predictable and easier to handle with consistency. The client had established a chain of events that we followed for each treatment regarding handing the dog off. The owner entered an already prepared room with the patient, then waited for a couple of minutes. The restraining technician entered the room first, to be handed the leash. The patient was asked to get up on the table and lie down sternal on the right side of the restraining technician. The treatment technician sat to the right of the patient. The patient was treated in a specific room, held with minimal restraint necessary for safety (right arm around neck, left hand holding the leash), and her vocalizations were ignored in favor of the technicians carrying on a normal conversation. Success was very dependent on the frame of mind the patient arrived in. All treatments were completed with only one therapy appointment involving increased vocalizing coupled with struggling against restraint consistently. All other appointments had infrequent and inconsistent vocalizing.

#### Case No. 2: Fearful Patient

A 5-year-old medium-sized mixed breed dog presented for back pain. The dog was not very treat motivated at home, and even less so when stressed. The client brought multiple types of treats to the consultation and used massage techniques to aid compliance. Chewy strips of sweet potato and peanut butter in a very thin consistency were readily consumed. He received laser treatments for 3 weeks, where he learned to relax.

When hydrotherapy was initiated, the patient had higher stress levels than he had demonstrated before, even at his consult. He would not take food. The underwater treadmill was a change in environment and expectation, and changes were stressful for him. He also had the added stressor of movement (whereas for laser therapy he had been relatively still, in a quiet room) along with being in a higher traffic, open-plan area. For three appointments, the technician and the client tried the same approach as was used in the consultation. The same treats, essential oils, and predictable massage were used while the patient walked in the underwater treadmill. Despite this, the patient demonstrated higher levels of stress than at the initial consult. Social pressure, in the form of paying direct attention, added stress for the patient; the owner repeatedly offered peanut butter for the patient to lick without expecting him to do so. The client and technicians continued to try offering a variety of treats, but stopped interacting with the patient purposefully while in the water. When the owner and technician had a conversation without paying attention to the patient, he was calmer. This was repeatable with future appointments. By the fourth underwater therapy appointment, the patient could walk calmly and take treats.

**Case No. 3: Food Allergy and Anxiety**

A 10-year-old Australian Shepherd with food allergies presented with biceps muscle pain and elbow osteoarthritis. The patient was already taking the medication fluoxetine for anxiety at initial presentation. During visits, she would shake and retreat into the corners of the room. Even light handling would result in a struggle to escape touch. Attempts were made to allow acclimatization, she visited just briefly for treats and then left. During these acclimatization visits, she was very nervous but when she took food, she relaxed. Treats used were dry kibble from home, supplemented with hypoallergenic treats in the clinic. After several desensitization visits, examinations were possible using this approach but treatment procedures (even the low stimulus of a modality) resulted in an escalation of her stress response. After several therapy sessions, she began to reach her threshold more quickly with each follow-up visit. Premedication was elected, using trazodone at home 2 hours before the car ride, but unfortunately had minimal effect. After communication with her primary care veterinarian, alprazolam was prescribed for situational use. This raised her threshold for stressors and enabled the team to complete rehabilitation.

The patient has some long-term issues, therefore her visits for follow-up at the clinic are very ordered with the aim of minimizing stress. She enters the room at least 10 minutes before the appointment time and listens to her favorite music from home (country and western from her owner's phone). With each visit, time is consciously allocated to talk about her progress at the start of the visit with no hands on for at least 5 minutes. The hypoallergenic treats given in clinic are not the same as her everyday home treats, therefore they provide higher motivation for luring and reward. Therapy decisions always take into consideration the stress of each plan and modality, and handling needed (e.g., contact needed along with beeping sounds of the laser versus soundless, no-contact pulsed electromagnetic field therapy).

**Case No. 4: Obese Patient with Orthopedic Disease**

A 35-pound (16 kg), 9-year-old Shetland Sheepdog (body condition score 8/9) presented for slow recovery from bilateral medial patellar luxation surgery. She was estimated to be 15 pounds (7 kg) overweight, due in part to two luxating patellas that did not allow her to fix her knees in extension and minimal activity for the past 6 months, but also due to being overfed for reduced activity. The rehabilitation veterinarian prescribed in-clinic therapy including hydrotherapy in the underwater treadmill. Initially the dog was not motivated to walk in the treadmill so the client used large pieces of freeze-dried liver to motivate and reward her (approximately 10 calories per average-sized treat). The dog would consume about half of her daily calories at each therapy session, while still being fed her usual meals.

The technician addressed the overfeeding by giving multiple options: use lower calorie treats, bring meals from home to use, or break the current choice into very small pieces. The client elected to use smaller pieces. An additional positive of using smaller "tastes" of treats is that the patient walked more consistently, with larger pieces she would stop walking to chew. The client was hesitant at first and it took a couple sessions to get the treats consistently small enough that the dog was only consuming about 20 calories per session. Motivation was maintained, while still being careful of total caloric intake.

#### Case No. 5: Home Environment Issues

A 6-year-old straight-line racing Whippet presented with a deltoid muscle strain. It was an acute injury that had occurred at an event. His owner had mobility problems and usually exercised her dogs to maintain fitness by letting them run in her large yard every day, chasing a Frisbee™. Practice on a lure line was once weekly but only in racing season. Visits for in-clinic therapy were not a problem, however the patient had to be exercise restricted at home, in a multi-dog household. A dog walker was utilized. In the clinic, appropriate modalities and manual therapies were used. Daily stretching was prescribed for home with consideration for the owner's disabilities. The patient was not making adequate progress in the rehabilitation plan, shoulder extension continued to be restricted, and some intermittent lameness was seen at a trot. After some discussion, it was determined that the owner had not been utilizing the dog walker daily and was tired of restricting her dog so had allowed some yard time for a mental break. The owner reported that the dog was much more relaxed, having been able to freely run, and was holding his front leg up only a little, a lot less than he had initially. She suggested she allow more activity and use an anti-inflammatory to help the pain. The pros and cons of this approach were discussed at great length and a compromise achieved. The patient could have off-leash yard time alone, not with the other dogs and no Frisbee.

The client agreed to make the long drive to the clinic twice weekly for exercise therapy for another month to bridge the gap between limited home exercise and full home exercise. The situation was not ideal, but the yard time alone progressed to Frisbee retrieves as the patient progressed, and then on to yard time with other dogs. An issue of lack of conditioning remains and straight-line racing fitness has not been achieved.

#### Case No. 6: Too Much Fun/Equipment Cues

A 7-year-old Rottweiler presented for strengthening and gait retraining. Previously he had been exercised on a land treadmill at home but it was overstimulating for him so had to be managed carefully. At his first underwater therapy session, he was put in a non-restrictive harness then led into the empty underwater treadmill. Since he seemed to be comfortable, although very excited, the treadmill started to fill with water. The patient began to thrash happily in the water. Attempting to hold him back with the harness only escalated his behavior more. His handler then remembered that a harness has only been used previously when doing bite sport work (a high-adrenaline and very reinforcing activity). He bit at, splashed, and tried to spin in circles in the water. He had never been exposed to water outside of a kiddy pool in the yard, to which his response had been similar. At this appointment, he was barely responsive to cues from his handler. He would not take food, and he could not be distracted by toys.

We developed a training plan to desensitize him to water. He had to perform obedience to earn access to the water. The sound of water splashing made it very difficult for him to perform the behaviors he was asked for. He had to demonstrate calmness to be able to stay in the water. We used a martingale lead to be able to prevent him from biting at the water if he was unresponsive to cues, and a calming cap (http://www.thundershirt.com/thundercap.html) to help him tune out external stimuli. His handler worked on his responses to his obedience position cues (sit, down, stand, walk forward specifically). Initially, he vocalized when not allowed to be in the water. After only two sessions in clinic, he could walk into and out of the underwater treadmill without trying to dig at or bite the low-level water sitting in there. He was responsive to cues from his handler. Homework included walking in and out of the edge of the water at a boat-loading dock. By the fourth session, the patient could walk in the moving treadmill in the water, although continued use of a calming cap was needed.

### Case No. 7: Motivating the Excitable Dog with No Focus

A 1-year-old Greater Swiss Mountain dog underwent rehabilitation therapy post surgery for a shoulder osteochondrosis. It soon became apparent that he lacked self-control. He was often overstimulated and could not focus when there was food present. It was nearly impossible to get thoughtful work with luring, as he was too focused on getting the food. He frequently stayed for the day so one technician was responsible for his therapy without help from the client. Teaching him how to earn food was the priority. He needed something to work on while walking in the underwater treadmill for strengthening, so he was taught a specific focal target in the treadmill. This was made easier because he would eat treats dropped in the water, so the technician could reward from any position by tossing the treat at the front of the underwater treadmill tank. The pieces of this final behavior that was shaped included walking in the water, head carried in a neutral position, eyes faced forward. It started with verbally marking, then rewarding a neutral head position with the technician standing in front. As this behavior became fluent, the technician began moving side to side in front, only marking and rewarding when the criteria were met. By the end of his treatment, he could walk for 20 minutes with reinforcement delivered on a variable schedule.

### Case No. 8: Motivating the Obese Feline Patient

RockinRhonda was a 13-year-old, calico, spayed, female, domestic shorthair cat that was painful, morbidly obese, and diagnosed with osteoarthritis. Her left stifle was worse than her right (both showed severe degenerative joint disease), and she had suspected previous bilateral cruciate ligament tears, plantigrade stance bilaterally in hindlimbs, and marked weight shift to front limbs (J Panko and J Bowra, personal communication, 2016, www.thespaw.com/).

The immediate rehabilitative goals were as follows:

- Stage 1: Assessment, pain management plan, activities of daily living
- Stage 2: Physical rehabilitation plan
- Stage 3: Lifestyle, fitness, core strength, mobility, and pain management lifelong care.

The equipment/modalities/therapies used included:

- Pain management
- Nutrition plan
- Joint health management
- Hydrotherapy
- Therapeutic laser
- Therapeutic exercise.

This morbidly obese kitty (22 lb (10 kg)) at initiation of rehabilitation is undergoing a plan that includes the following:

#### *Pain Management and Joint Health*

- Onsior® (robenacoxib 6 mg orally once daily), evaluating efficacy daily and reducing dose in correlation with weight loss.
- Therapeutic laser (see Stage 2: Physical rehabilitation plan).
- Glycoflex Chews™ (loading does of 3 chews per day for 6 weeks) then 2 chews daily, reducing dose in correlation with weight loss.
- Cartrophen® injections (0.33 mL subcutaneously every 7 days for 4 doses, then one dose every 30 days, reducing dose volume per weight loss).

#### *Nutrition Plan*

- Assess body fat index and create a safe weight loss and management plan using the Hill's Healthy Weight Protocol™.
- Monthly weigh-ins. Recalculate nutrition plan monthly.
- Measured and controlled Hill's Metabolic® Wet and Dry + Glycoflex® chews.

*Physical Rehabilitation Plan*

- Low-impact exercise sessions (underwater treadmill) 2–3 times daily for up to 5 minutes, increasing length of sessions and reducing buoyancy as tolerated.
- Swimming in pool to improve cardiovascular fitness and assist with weight loss.
- Once she can tolerate it, progressively add in cavaletti poles flat on the ground, progressing to 5 cm then 8 cm off the ground to promote front and hind limb weight shifting, flexion, and extension, weight shifting with front limbs on balance pad, progressing to front limbs on balance disc, progressing to having front limbs on yoga block while eating to increase challenge of hind end weight shift.
- Environmental modifications: Provide low litter box with 3 cm step in, padded bedding, and medium-sized dog kennel. As hind end strength improves, raise feed bowls to promote weight shifting to hind end and engagement of core muscles while eating and return to normal litter box use to promote flexion of stifles and hips.
- K-Laser™ stifles, elbows, and shoulders, 3 watts, 1 cycle each joint, twice weekly for 3 weeks, then re-evaluate. Move to weekly laser sessions when mobility improves.
- As pain and weight diminish, incorporate therapeutic exercise program to increase core strength, build hind limb muscle mass, improve hind limb strength and plantigrade stance, overall mobility, and ability to complete activities of daily living.
- Monthly re-evaluations with Dr. Jeff Bowra, DVM, CCRP or, as concerns arise, progress stops, or plan needs modification.

*Long-Term Lifestyle and Mobility Plan*

- Ongoing pain management, weight management, joint health management and physical rehabilitation with Jenn Panko, RVT/CCRP, OCMC, CAPMC, under the supervision and direction of Dr. Jeff Bowra, DVM, CCRP.

At the current time (August 25, 2016) RockinRhonda is 11 lb (5 kg) and is continuing with her prescribed physical rehabilitation plan.

## Motivating the Client

Motivating the patient is always the most necessary step for in-clinic rehabilitation. For home rehabilitation, we have two factors – we must motivate the client to motivate the patient. Daily home exercises (often several times a day) can become difficult to fit into a busy schedule, and therefore knowing the importance of the prescribed exercises along with the reasoning for each exercise can help to motivate the client (and every person involved in home care of the patient). Some clients want to know exactly why the prescribed exercises will help, others just want to know that the exercise will help. Explaining the exercise verbally when first introducing it, and explaining in written handouts and videos helps to aid motivation. Demonstrating the exercise with the patient shows the client that the patient is willing to perform the exercise. The next hurdle is to use the client to motivate the patient. This can be a significant challenge. It helps to meet all members of the family as there may be one member who is particularly adept at motivating and working with the pet. This may not be the primary caregiver. Sometimes the client's compassion for the patient may present a roadblock in that the caregiver does not want to hurt their pet. Explaining how the exercise should feel and the relative "cost" (slight initial discomfort during a stretch for example) versus benefit can help to motivate the client to perform the exercise. As stated earlier, no exercise should be truly painful; in the veterinary world we rely on patient compliance and a breakdown of this compliance can mean an end to therapy. Motivating your patient and client in a low-stress way can achieve great goals.

## Conclusions

A behavior-centered rehabilitation clinic will be a successful clinic. Motivating a patient with positive (usually food) rewards in a calm environment aids in a low-stress visit for examination, and subsequently for therapy. As the number of visits increases, the stress level of the patient should ideally decrease, or at least not escalate. Minimizing handling and other stressors is vital where possible. A compliant, relaxed patient will benefit greatly from rehabilitation therapy. Home compliance for exercises is an important part of therapy, and motivating both the patient and the client leads to a successful outcome and a rewarding experience with rehabilitation. The trust gained from a low-stress therapeutic course for a patient and client will stand the practice in good stead for reaching therapeutic goals, and for future referrals.

## References

American Association of Feline Practitioners (2004) Pleasant veterinary visits for cats. In *Feline Behavior Guidelines*. www.catvets.com/public/PDFs/PracticeGuidelines/FelineBehaviorGLS.pdf, p. 20 (accessed March 9, 2016).

Martin KM and Martin D (2009) How dogs learn. In *Puppy Start Right: Foundation Training for the Companion Dog* (eds. KM Martin and D Martin). Sunshine Books Inc., Waltham, MA, p. 28.

Overall KL (2013) Changing behavior, roles for learning, negotiated settlements and individualized treatment plans. In *Manual of Clinical Behavioral Medicine for Dogs and Cats* (ed. KL Overall). Elsevier Mosby Publishing, St. Louis, MO, p. 76.

Pavlov IP (1928) *Lectures on Conditioned Reflexes* (transl. WH Gantt). Allen and Unwin, London.

Skinner BF (1938) *The Behavior of Organisms: An Experimental Analysis*. Appleton-Century, New York.

## Resources

Through A Dog's Ear. www.throughadogsear.com

Understanding canine body language. http://www.doggiedrawings.net

### Interactive feeding toys for dogs

Busy Buddy Toys. http://www.petsafe.net/busybuddy

Green slow feeder. http://northmate.com/category/products/green/

Kong. www.kongcompany.com

Orka. http://www.petstages.com/our-products/

### Interactive feeding toys for cats

Slimcat.http://store.petsafe.net/slimcat-interactive-feeder

Trixie Fantasy Board. https://www.trixie.de/heimtierbedarf/en/shop/Cat/

### Interactive feeding toys for horses

Amazing Graze. www.horsemenspride.com/products/#tour-5